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장소 | 청풍리조트 레이크호텔 컨벤션홀

일자	시간	프로그램
1/26(월)	14:00~14:10	개회
	14:10~15:40	테마 주제 강연
	15:40~16:00	포스터 홍보 시간 I
	16:00~17:30	포스터 발표 I
	17:30~18:00	체크인
	18:00~19:00	저녁 만찬
	19:00~20:30	소셜 네트워킹
1/27(화)	07:30~08:30	조식
	09:00~09:20	포스터 홍보 시간 II
	09:20~10:50	포스터 발표 II
	11:00~11:30	시상 및 폐회



한국뇌신경과학회

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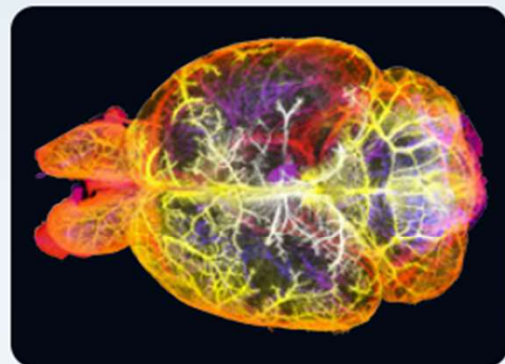
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Coordination among Putamen, vmPFC, and FPN Supports Sequential Rule Generalization in Unspecified Environments

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Generalization is traditionally attributed to high-level control regions such as the frontoparietal network (FPN). While perceptual generalization is well documented, the mechanisms by which humans generalize latent sequential rules remain elusive. This process requires internal representations of abstract structure, which the ventromedial prefrontal cortex (vmPFC) is thought to encode as cognitive maps. However, it remains unclear whether these representations, grounded in prior knowledge, support sequential rule generalization. Furthermore, it remains poorly understood how humans arbitrate among competing strategies in environments where multiple plausible rule structures can be inferred. The putamen is thought to play a critical role in evaluating alternative reorganizations of prior knowledge. Here, we test the hypothesis that sequential rule generalization in such environments relies on the orchestrated coordination among the putamen, vmPFC, and FPN. We designed a cue-based sequence learning task in which cues could map onto rules in multiple ways. We compared participants pretrained on rules for only a subset of cues (experimental group) to those without prior rule knowledge (control group). Behaviorally, the experimental group adopted the optimal strategy more rapidly. fMRI results revealed sustained FPN engagement throughout generalization, whereas putamen activation increased with learning. Successful generalization was marked by enhanced putamen–vmPFC functional coupling, with the vmPFC representing the optimal rule structure as learning unfolded. These findings indicate that sequential rule generalization in unspecified settings relies on a coordinated system: the putamen evaluates strategies, the vmPFC forms progressively updated abstract rule representations, and the FPN exerts cognitive control. This study highlights a neural mechanism integrating valuation, structure representation, and cognitive control to infer abstract sequential structure.

Keywords: Generalization, Sequence, fMRI

Acknowledgements:

Robust pose tracking and behavior analysis pipeline for phenotypically similar multi-animal interaction

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Recent advances in deep learning-based animal pose estimation and tracking have enabled quantitative analysis of complex animal behaviors in neuroscience research. While pose estimation for individual animals has achieved high accuracy, current methodologies for multi-animal tracking in freely moving settings remain constrained by limitations. These include computational costs and inference times, as well as reduced tracking quality due to ID-switches caused by phenotypic similarities among animals. To address these limitations, we introduce MovAl (Move-Altogether), a novel behavioral analysis pipeline for accurate and efficient tracking of multiple freely moving animals. MovAl significantly improves animal tracking accuracy by applying a color-coded preprocessing step to each instance using a video object segmentation (VOS) method that leverages a strong memory attention. As a result, MovAl reduces computational cost and inference time while increasing pose accuracy and decreasing ID-switch errors, which are critical for behavioral analysis, compared with other widely used pipelines. Furthermore, we developed a new autoencoder architecture to analyze behavioral patterns of three mice freely moving in a home-cage environment. This approach revealed rapid, age-dependent changes in behavioral patterns in adolescent mice between 3 and 5 weeks of age, with differing rates of change between males and females. We additionally adopted a language-model framework and found that the rules governing behavioral patterns undergo rapid reorganization during this developmental stage. Finally, we applied MovAl to various experiments, including object exploration and mom-pup interaction, to validate the robust tracking and to establish its utility as a versatile tool for freely moving behavior. Together, these results demonstrate that MovAl provides a robust and biologically informative framework for high-precision multi-animal behavior analysis across diverse experimental setting.

Keywords: Video Object Segmentation, YOLO, Behavior patten analysis

Acknowledgements: RS-2024-00456343, RS-2025-02303740

Projection-specific diversity of dopaminergic activity under aversive situations

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Dopamine signals in the brain have been widely implicated in reinforcement learning, particularly through encoding reward prediction errors (RPEs) as formalized by temporal-difference (TD) learning models. Mesolimbic dopamine activity, such as that in the ventral striatum (VS), has been shown to conform to a derivative-like form of a value function in both rewarding and aversive contexts throughout experiments. Meanwhile, the tail of striatum (TS) is the region that is recently known to play an important role in threat prediction. However, the computational basis has yet to be fully understood. We designed an active avoidance task using virtual reality in which mice learned to avoid receiving an electric shock by running out of a specific zone. Fiber photometry recording from VS in the previous work revealed that dopamine signals decreased at the entrance of the aversive zone and increased after they got out of the zone ($n = 14$ mice). Our probe experiments, including speed gain changes or teleportation in a virtual linear track, supported the TD RPE hypothesis ($n = 8-13$ mice). In contrast, TS dopamine signals exhibited a markedly distinct pattern: it showed gradual increase of dopamine signal at the entrance of the shock zone and dip down at the end of it ($n = 5$ mice). Each shock reliably induced a phasic excitation in TS dopamine levels. We next designed a bar-moving protocol, which shows a horizontal bar moving down from the top and the animal receiving electric shock when the bar reaches the end. The TS dopamine ramped up as the bar moved down to get closer to the aversive stimuli, resembling the VS dopamine getting closer to the reward ($n = 4$ mice). Together, our results highlight the regionally distinct pattern of striatal dopamine circuits in shaping adaptive behavior under aversive conditions, providing a further understanding of how the brain encodes negative value through dopaminergic pathways.

Keywords: Avoidance, Dopamine, Striatum

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Differential encoding of value-based contextual behavior in the intermediate and dorsal regions of the hippocampus

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The cognitive map in the hippocampus may represent both spatial and non-spatial information. Prior studies showed that the intermediate hippocampus (iHP) exhibits stronger value-dependent place coding than the dorsal hippocampus (dHP) during value-based navigation (Jin and Lee, 2021; Jin et al., 2024). However, given that the iHP has broader firing fields and fewer place cells than the dHP, its function may extend beyond encoding a specific location's value. Instead, the iHP may contribute to value-dependent navigational decision-making (e.g., left or right) based on the spatial context. To test this, we simultaneously recorded single-unit spiking activity from the dHP and iHP, while body-fixed rats (male, Long-Evans; n=5) performed a contextual memory task in a VR environment. Rats were pseudorandomly placed in one of two spatial contexts (A or B), running along a 2-m linear track. Once passing an invisible choice line on the track, two lick ports (left or right) were extended near the animal's snout. The rats were required to lick one of the ports providing a higher-value reward (40 μ l sugar water) in each context (i.e., left and right ports in context A and B, respectively) and avoid the port associated with a lower-value reward (5 μ l). We also reversed the context-choice value contingencies in the middle of the session. Rats successfully learned the task, including contingency reversals ($\geq 70\%$ high-value-port-licking). Preliminary data showed that both dHP and iHP neurons exhibited robust spatial context selectivity. Interestingly, several iHP contextual neurons showed little location-selectivity, whereas most dHP contextual neurons displayed strong location-selectivity. Notably, contextual neurons in the iHP, but not in the dHP, drastically altered their firing patterns following reversal and fired more strongly during high-value-port-licking trials than low-value-port-licking trials. These results suggest a role for the iHP in value-dependent contextual behavior.

Keywords: Intermediate hippocampus, Value-based navigation, Spatial decision-making

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Pharmacological Activation of REV-ERB α Attenuates Dopamine Dynamics and Reduces Cocaine-Seeking Behavior

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Drug addiction is associated with neuroadaptations in the brain's reward system, particularly dysregulated dopamine signaling. Because this dysfunction reflects altered regulation of the dopamine system, rather than moment-to-moment changes in neurotransmitter release, there is a need for molecular approaches that act upstream of synaptic transmission. Recent studies have shown that REV-ERB α transcriptionally represses tyrosine hydroxylase (TH), the rate-limiting enzyme in dopamine synthesis, thereby acting as a critical negative regulator of dopamine levels. Given this role, we hypothesized that targeting REV-ERB α could modulate dopamine dynamics and attenuate addictive behavior. Using fiber photometry with a genetically encoded dopamine sensor (GRABDA2m), dopamine (DA) activity in the nucleus accumbens (NAc) was measured during a sucrose lick test. We found that DA transients following reward delivery were significantly attenuated by a REV-ERB α agonist (STL1267), whereas a REV-ERB α antagonist (SR8278) significantly increased DA release. To determine whether bidirectional modulation of dopamine dynamics translates into changes in addictive behavior, cocaine self-administration (0.25 mg/kg/infusion) was established under fixed-ratio (FR) 1, FR3, and FR5 reinforcement schedules during daily 2-h sessions over 14 days. Thereafter, the REV-ERB α modulator was microinfused into the ventral tegmental area (VTA) 3 hours before sessions in which lever presses no longer resulted in cocaine delivery. Agonist treatment significantly accelerated the extinction of cocaine-seeking behavior, whereas antagonist treatment had no effect on extinction, with response levels remaining similar to control conditions. Taken together, our findings indicate that molecular modulation of dopamine synthesis via REV-ERB α effectively regulates reward-related dopamine signaling and suppresses addictive behavior, identifying REV-ERB α as a promising therapeutic target for addiction.

Keywords: Addiction, REV-ERB α , Dopamine

Acknowledgements:

Representation of spatial information by single units in the non-human primate (NHP) hippocampus during spatial navigation in virtual environment

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The hippocampus plays a critical role in spatial navigation and memory. In non-human primates (NHPs), studies have commonly identified spatial view cells, which fire in response to gaze directed toward specific parts of the environment, whereas in rodents, place cells are more well established. This species-specific divergence has led to ongoing debate about whether canonical place cells, as defined in rodents, also exist in the primate hippocampus, or whether spatial representations in NHPs rely on fundamentally different coding principles. To address this question, we recorded single-unit activity from the hippocampus of two NHPs while they performed an object-in-context memory task in a virtual reality (VR) environment. In each lap, the NHP entered either a forest or city context and navigated along a linear track with a U-turn rotary using a joystick. During both outbound and inbound navigation, the NHPs were stopped at predefined trial locations to perform a contextual object choice trial. To investigate spatial representations specifically linked to navigation, we focused our analyses on the segments during which active navigation occurred—when the NHPs were moving through the spatial context. Among the recorded single units ($n = 120$) in the hippocampus, we observed both canonical place cell and spatial view cell activity. A total of 42.5% ($n = 55$) of units qualified as place cells. Similarly, 44.7% ($n = 55/114$) of units met the criterion for putative spatial view cells. Notably, a subset of cells encoded both spatial location and view information (60%, $n = 39$), while others were selectively tuned to view (18.5%, $n = 12$). Importantly, approximately 21% ($n = 14$) of spatially responsive cells exhibited place-specific activity without significant correlation to view, suggesting location-based firing independent of gaze direction. These results suggest that the primate hippocampus integrates both visual and positional information to support spatial memory.

Keywords: Hippocampus, primate, place cell

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Insight in rats: Behavioral investigation in Tolman’s Maze

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When faced with complex problems, both humans and animals can display sudden, unexpected shifts in behavior, commonly interpreted as evidence of “insight.” First reported nearly a century ago in Tolman and Honzik’s maze study, such abrupt reconfigurations in problem-solving strategies have long been taken as evidence that animals can reorganize internal representations of the task. Despite its historical importance, however, the mechanistic nature of insight remains elusive, partly due to limitations in behavior quantification. Our primary goal is to identify the precise temporal moment at which such a transition occurs. We define insight as the discovery of an optimal problem-solving strategy that is not explicit in the task, triggered by the sudden integration of previously underweighted task variables. To this end, we revisit this classic paradigm by constructing a behavioral re-implementation of Tolman’s maze and using it as a platform to examine insight-like problem-solving in rats, while incorporating an AI-based behavior analysis to systematically characterize behavior across task experience. Rather than emphasizing gradual changes in aggregate performance, we conceptualize insight as a discrete shift in problem-solving strategy, expressed as a sudden change in behavioral outcomes across trials. By grounding the study of insight in a behavioral reenactment of a foundational paradigm, this work clarifies the experimental parameters of “insight” and provides a concrete approach for spatiotemporally locating its occurrence in the brain. This perspective establishes a foundation for future studies seeking to link sudden cognitive transitions to their underlying neural mechanisms—such as innovative neural patterns or specific biomarkers—and contributes to a deeper understanding of high-level cognitive phenomena such as flexibility, creativity, and problem-solving.

Keywords: Insight, Problem-solving, Tolman’s maze

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Distinct Roles of Motor and Thalamic Inputs for Egocentric Spatial Coding in the Retrosplenial cortex:

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Effective navigation requires not only encoding the current spatial configuration of the environment but also predicting future states based on ongoing movements and head orientation. The retrosplenial cortex (RSC) is a key hub for egocentric spatial representations and is anatomically interconnected with motor-related regions, such as the secondary motor cortex (M2), and head-direction circuits in the anterior thalamic nuclei (ATN). However, whether and how RSC neurons encode egocentric information in a predictive manner, and how motor- and head-direction-related inputs contribute to this process, remain unclear. To address this, we performed in vivo calcium imaging of excitatory neurons in the RSC of mice navigating a spatial environment. We quantified egocentric boundary tuning across temporal offsets between neural activity and behavior to assess predictive encoding. We found that a subset of RSC neurons exhibited prospective encoding of egocentric angles, with neural activity preceding behavioral states, independent of movements. To examine the circuit mechanisms underlying this predictive coding, we selectively inhibited inputs from ATN or M2 using optogenetic silencing. ATN inactivation significantly disrupted egocentric tuning strength and temporal alignment, whereas M2 inactivation selectively reduced predictive encoding during turning behavior without abolishing baseline egocentric representations. These results demonstrate that RSC neurons encode egocentric spatial information in a predictive manner and that motor and thalamic inputs play distinct and complementary roles in shaping this computation. Together, our preliminary result suggest that predictive egocentric coding in the RSC emerges from the integration of movement-related and head-direction signals within distributed navigation circuits.

Keywords: Retrosplenial cortex, Egocentric representation, Calcium imaging

Acknowledgements:

Sequential Learning of Stimulus Mappings and Rules Revealed by Time-Resolved EEG Decoding

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Living organisms excel at sequential learning, the ability to form internal cognitive maps through experience and flexibly apply them to novel environments. Understanding the neural mechanisms is crucial for uncovering the basis of higher intelligence. We designed a rule-based object-to-position association task to investigate how the human brain constructs and utilizes cognitive maps during sequential learning. Participants performed a spatial decision-making task while 64-channel electroencephalogram (EEG) data were recorded. In each trial, subjects fixated on a central point, briefly viewed an image, and selected its associated location from five options arranged in a pentagon. During an intermediate delay, a color cue (red, blue, or yellow) indicated the rule: red for standard mapping, blue for a 72-degree clockwise rotation, and yellow for a 144-degree rotation, necessitating flexible adjustment of learned spatial associations. Behavioral results showed initially low accuracy that improved significantly with practice, reflecting cognitive map formation. The introduction of a new rule (blue cue) temporarily reduced accuracy, followed by rapid recovery. Notably, when a third rule (yellow cue) was added, performance remained stable, suggesting successful generalization of prior knowledge to novel conditions. To identify neural correlates of these adaptive behaviors, we developed a spatial position classifier using support vector machines and applied it to EEG data. Temporal generalization analysis revealed that location-specific neural representations emerged after image presentation and were flexibly updated following rule cues. These findings demonstrate that participants formed stable image-location associations and dynamically transformed them according to rule changes, as evidenced by EEG activity patterns. This study provides novel insights into the cortical dynamics of sequential learning and the neural basis of cognitive map formation and flexibility.

Keywords: Sequential learning, Electroencephalogram, Temporal generalization analysis

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Complementary Roles of Theta and Alpha Waves in the Cortical-hippocampal Network during Episodic Memory Encoding Using MEG

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Previous studies have highlighted the role of hippocampal theta oscillations (2-8Hz) in sequential episodic binding, while cortical alpha oscillations (8-14Hz) have been implicated in load-dependent information processing. Despite their shared relevance for memory performance, how these oscillations interact to guide human memory function remains unclear.

Here, we investigated cortical-hippocampal oscillatory dynamics using MEG with source reconstruction in six epilepsy patients (ages 13-29) recruited from Yonsei Severance Hospital, while they performed a visual object-space-time binding episodic memory task. We observed pronounced alpha and theta oscillations along the scene-memory pathway from visual cortex to hippocampus.

Spectral power analyses in each brain region showed opposing influences of theta and alpha on memory binding across task phases: stimulus-evoked theta power in the early visual cortex was positively correlated with memory, while inter-stimulus alpha power in the medial temporal lobe was negatively correlated. This pattern suggests distinct roles of theta in memory encoding and alpha desynchronization in memory retention.

Increased theta synchrony within the hippocampal region and reduced alpha synchrony between the hippocampus and cortex were associated with successful episodic memory binding. These results indicate that, along with robust theta-modulated hippocampal memory processes, alpha oscillations play a regulatory role between the cortex and hippocampus, especially during the inter-stimulus period when memory retention is critical for successful memory.

In conclusion, alpha and theta oscillations play complementary roles in episodic memory binding: synchronized theta in the cortical and hippocampal regions supports memory representation during encoding, whereas desynchronized alpha functionally separates cortical and hippocampal regions, potentially preventing external input from interfering the memory representation.

Keywords: Cortical-hippocampal Network, Theta and Alpha, Episodic Sequence Encoding

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Bridging Anatomical and Effective Hierarchies via Neural Mass Modeling

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The cerebral cortex is organized into a hierarchical structure defined by laminar-specific connectivity, quantitatively estimated using the fraction of Supragranular Labeled Neurons (SLN). Conventionally, this structure is assumed to dictate a functional "counterstream" architecture, where feedforward pathways are predominantly excitatory and feedback pathways are inhibitory. However, this rigid assumption often fails to account for the complex dynamics of large-scale brain networks. Here, we interrogate this relationship using a biologically informed neural mass model with trainable long-range excitatory (LRE) and inhibitory (LRI) weights, decoupling functional effects from the fixed structural scaffold. We demonstrate that a "Fixed-SLN" model, which strictly enforces the canonical rule, fails to reconstruct empirical dynamics, indicating that strictly deriving functional weights from anatomy is insufficient. In contrast, optimizing weights to capture effective connectivity (EC) spontaneously recovers the latent counterstream architecture without anatomical priors. Furthermore, we observed the recruitment of feedforward inhibition to stabilize strong driving inputs. The resulting "Effective Hierarchy," reconstructed from these optimized weights, correlates with but significantly deviates from the anatomical hierarchy. These findings suggest that the canonical rules of feedforward-excitation and feedback-inhibition are not fixed constants but dynamical necessities—flexible mechanisms that emerge to stabilize information flow within the structural scaffold.

Keywords: Structure-Function Relationship, Effective hierarchy, Neural mass model

Acknowledgements:

Behavioral and neural correlates depending on emotion-valence during social affective recognition in mice

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Empathy, the ability to recognize and interpret the emotional states of others, is crucial for appropriate social interactions and often impaired in psychiatric disorders. Although prior studies have primarily focused on negative emotions such as fear and pain, how animals recognize and distinguish a broader range of emotions, including positive ones, remains unclear. Here, we asked whether mice merely detect emotionally salient social cues or actively discriminate the emotional valence of conspecifics. First, we examined behavioral responses and neural activation following home cage interaction with stressed or relieved. Despite the comparable levels of overall social interaction, mice displayed valence-specific behavioral patterns, accompanied by distinct c-Fos activation across brain areas. In the emotion discrimination test (EDT), mice preferentially investigated stressed or relieved conspecifics over neutral controls, consistent with previous findings. However, when presented with both stressed and relieved conspecifics, mice approached stressed conspecifics more rapidly while spending comparable interaction time with relieved conspecifics. Unbiased clustering analysis of EDT behavioral revealed emotion-specific behavioral patterns. Together, these results demonstrate that mice discriminate the emotional valence of social partners and engage distinct neural networks during social affective recognition.

Keywords: Emotion recognition, Social interaction, Unbiased clustering

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Common neural processes of self-distanced memory retrieval across diverse memory contents

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Beyond the conventional first-person perspective, autobiographical memories can be retrieved through self-distancing – an observer-like viewpoint that stands in contrast to the more common, self-immersed mode of recall. Previous studies demonstrate that self-distancing alters the content of retrieved memories, suggesting distinct underlying neural process. To address this, we conducted an fMRI study incorporating two distinct phases: a pre-scan interview and a subsequent retrieval session. During the pre-scan interview, we collected vivid autobiographical memories of each participant, which were subsequently recalled inside the MRI scanner under both self-distancing and self-immersion conditions. We found that self-distanced retrieval, compared to self-immersed retrieval, elicited increased mean activation in the middle frontal gyrus (MFG), inferior frontal sulcus (IFS), angular gyrus (AG), and precuneus. Focusing on these regions, we subsequently investigated whether self-distancing recruits shared neural processes by comparing pattern similarity in neural responses both within and across participants. Notably, within individual participants, self-distancing – relative to self-immersion – yielded significantly greater shared neural representations across trials, regardless of memory content. This consistency was more pronounced in the left hemisphere than the right hemisphere. Furthermore, shared representations in the MFG and precuneus persisted even across different individuals during the self-distanced condition. These findings suggest that adopting a self-distanced perspective during retrieval engages common neural process across diverse memory contents and individuals, particularly within prefrontal and parietal regions.

Keywords: self-distancing, autobiographical memory, fMRI

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Dissociative Neurocognitive Effects of Social Stress on Object-Space Memory in Adolescents with Mild and Severe Depressive Symptoms

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Object-space association in episodic memory is widely known to depend on the hippocampus. While stress is a critical factor influencing memory function, most research has been focused on the negative effects of stress on the hippocampus. However, responsiveness to stress is highly variable across individuals and may interact with their affective traits. Moreover, such effects of stress on memory and cognition may be pronounced during adolescence, a period of heightened stress sensitivity and variability in mental health. An associative memory task was conducted in two independent experiments (Exp. 1: N=62, mean age 15.5; Exp. 2 with fMRI: N=57, mean age 16.2), in which participants encoded objects encountered while navigating a virtual maze and were subsequently tested on object-scene association. Social stress was applied in a condition in which adult observers appeared on the screen with negative feedback on performance. The CES-D scale measured self-reported depression symptoms. In Exp. 1, higher depressive symptoms were associated with worse object-scene memory; but interestingly, adolescents with mild depressive symptoms showed enhanced object-scene memory under social stress. Exp. 2 was conducted in the fMRI to find the neural correlates of this stress-induced memory modulation. In the no stress condition, greater activity in the anterior hippocampus and anterior cingulate cortex (ACC) were associated with better memory. Under social stress, high depression levels were associated with reduced hippocampal activity and worse memory. However, mild depression levels were associated with enhanced connectivity between the amygdala and ACC, which was related to improved memory. We speculate that social stress induced a positive effect on memory in otherwise healthy adolescents with mild depressive symptoms due to improved allocation of attention to the spatial scene, reflected in the recruitment of the ACC-amygdala salience network function.

Keywords: Object-space memory, Adolescence, Social stress

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Modulation of Capsaicin Value Based on Internal States in Mice

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Internal physiological states play a central role in shaping how organisms evaluate and respond to environmental stimuli. Previous work has shown that internal states such as hunger or thirst modulate reward processing, particularly within dopaminergic systems. However, it remains unclear whether internal state can modulate the functional value of aversive nociceptive stimuli during learning and valuation, rather than merely scaling their sensory unpleasantness. To address this question, we examined how thirst state influences behavioral and dopaminergic responses to capsaicin in mice. Behaviorally, mice exhibited low voluntary intake of capsaicin relative to water, necessitating experimenter-controlled exposure paradigms to reliably assess dopaminergic responses. Using fiber photometry to measure dopamine activity in the nucleus accumbens core, we found that dopaminergic responses to capsaicin were strongly modulated by internal thirst, both within and across days. Under thirst conditions, capsaicin consumption elicited dopaminergic excitation, whereas under water-satiated conditions, dopaminergic activity was shifted toward inhibition. Together, these findings demonstrate that dopaminergic responses to an aversive stimulus are dynamically shaped by internal physiological state, supporting a state-dependent assignment of functional value beyond sensory unpleasantness. Whole-brain activity mapping using 15.2 T functional magnetic resonance imaging (fMRI) and homeostatic reinforcement learning modeling are currently ongoing to further characterize the neural and computational mechanisms underlying state-dependent valuation.

Keywords:

Acknowledgements:

Map-like Representations of Social Recognition Dimensions in the Human Hippocampus

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Recognizing social relationships is fundamental for understanding and interacting within complex social environments. Our prior work has identified two principal dimensions underlying interpersonal relationship recognition: Amity-Hostility dimension (PC1) and Restrained Amity-Suppressive Hostility dimension (PC2). These dimensions are processed in distinct cortical regions, with PC1 predominantly encoded in the right posterior superior temporal sulcus (rpSTS) and PC2 primarily represented in the right ventromedial prefrontal cortex (rvmPFC). However, the mechanism by which these dimension-specific signals are integrated into a unified representation of interpersonal relationships remains unclear. In the present study, we investigated whether the hippocampus encodes social relational information within a two-dimensional space defined by the PC1 and PC2 dimensions, analogous to spatial mapping. To address this, we conducted an event-related functional magnetic resonance imaging (fMRI) experiment. During the experiment, participants viewed naturalistic movie clips and evaluated the social relationship between a designated actor and target character. Representational similarity analysis (RSA) and support vector regression (SVR) revealed that the hippocampus represents interpersonal relationships based on distances in the PC1-PC2 space. In contrast, the rpSTS and rvmPFC selectively encoded dimension-specific information corresponding to the PC1 and PC2 dimensions, respectively. Mediation analysis indicated that hippocampal representations reflect distinct contributions of the rpSTS and rvmPFC. Feature transfer analysis further demonstrated unidirectional information flow from cortex to hippocampus, with no evidence for reverse transfer. Collectively, these results suggest that the hippocampus integrates dimension-specific cortical inputs into a unified, map-like representation of interpersonal relationships.

Keywords: Hippocampus, Social Recognition, fMRI

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The physiological and functional characterization of a novel population of VGluT3+ neurons in the aBNST

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Vesicular glutamate transporter 3 (VGluT3) is often co-expressed with other vesicular transporters and supports atypical, region-dependent co-transmission. However, whether VGluT3-expressing neurons share conserved properties across brain areas or adopt region-specific identities remains unclear. The bed nucleus of the stria terminalis (BNST) contains a high density of VGluT3-expressing neurons (VGluT3+ neurons); however, the cellular mechanisms and functional relevance are poorly understood. Here, we characterized VGluT3+ neurons in the anterior BNST (aBNST). Transcriptomic meta-analyses revealed that aBNST VGluT3+ neurons form a molecularly distinct population, differing from both neighboring VGluT3 neurons and VGluT3+ neurons in the dorsal raphe nucleus. Electrophysiological characterization further demonstrated unique intrinsic membrane properties. Functionally, aBNST VGluT3+ neurons project locally within BNST targets and to the paraventricular nucleus, where they predominantly provide inhibitory synaptic input, while co-releasing glutamate and GABA with asymmetric dynamics. Notably, food deprivation selectively increased voltage sag in these neurons, whereas acute restraint stress had no effect. Together, these findings identify aBNST VGluT3+ neurons as a regionally specialized co-transmitting population selectively engaged in feeding-related circuit modulation.

Keywords: Bed nucleus of stria terminalis, VGluT3, Electrophysiology

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Spatial boundaries modulate sequential episodic memory with age-related differences

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Spatiotemporal binding, relative to object-related information, is both robust in adult episodic memory and emerges earlier in development. Given the role of space as a framework for binding elements of episodic memory, a boundary-based representation of environmental geometry may also provide a stable anchor for episodic memory.

Building on these findings, we investigated whether introducing a spatial boundary alters the event structure of sequential episodic memory, with a focus on the medial temporal lobe, and examined how such processes are affected by aging.

During fMRI acquisition, participants (Age=19-78; N=182) watched objects being placed into baskets distributed throughout a virtual room and reenacted the episode by selecting objects and their corresponding locations in order. In the boundary condition, a spatial boundary divided the room into two sides, such that the event sequence unfolded across both sides.

First, we found that young adults showed higher spatiotemporal memory accuracy in the boundary condition, particularly for events occurring after the boundary crossing. The parahippocampal cortex (PHC) showed greater activation in the boundary condition, and the between-condition difference in PHC activity was positively associated with boundary-related space-time binding enhancement. Although our task did not involve an explicit perceptual shift during crossing, the PHC and hippocampus showed increased responses when the event sequence crossed the spatial boundary. As in the young group, older adults showed higher space-time accuracy in the boundary condition. Notably, the boundary impact varied with age - beneficial for space-time yet disruptive for object-time binding.

These findings suggest that spatial boundaries can elicit event segmentation, helping to restructure temporal memory through the PHC and hippocampus. Furthermore, this space-time binding enhancement is robust across aging, while object-time binding in episodic memory declines.

Keywords: Episodic memory, Aging, fMRI

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Repeated stress exposure impairs social- and aversion-induced dopamine signaling in distinct nucleus accumbens subregions

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The nucleus accumbens (NAc) integrates rewarding and aversive stimuli through dopaminergic input from the ventral tegmental area (VTA), yet how acute stress alters this signaling across NAc subregions remains unclear. Here, we investigated the impact of acute learned helplessness (aLH) on dopamine dynamics in the lateral and medial NAc shell (NAcLsh and NAcMsh). Ex vivo patch-clamp recordings showed increased neuronal excitability selectively in NAcMsh neurons after aLH, and this increase was abolished by a D1 receptor antagonist, suggesting dopamine-dependent plasticity. To examine subregion-specific dopamine dynamics in vivo, we recorded dopamine release using GRABDA sensors during social interaction, reward consumption, and foot shock. Behaviorally, aLH reduced social interaction and mobility during aversive stimulation. Social interaction triggers dopamine release in the NAcLsh, which was significantly attenuated following aLH, whereas dopamine release in the NAcMsh was selectively evoked by foot shock and similarly diminished by aLH. Consistent with these functional differences, retrograde tracing revealed largely distinct VTA populations projecting to each subregion. These findings demonstrate that acute stress disrupts dopaminergic signaling in the NAc shell in a subregion-specific manner, impairing both social and aversive processing. Such disruptions may underlie broad motivational deficits observed in stress-related disorders.

Keywords: Nucleus accumbens, Dopamine, Stress

Acknowledgements: RS-2022-NR072375

Dopamine activity in the tail of striatum predicts avoidance behaviors in complex threatening situations

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Previous animal studies have commonly used discrete sensory cues to associate fear. However, in humans, fear memories are often triggered by complicated visual elements. Using more naturalistic visual stimuli would enable us to investigate how fear is processed in real life of humans. We developed an active avoidance task in augmented reality (AR) where walls of a virtual room with variable patterns are presented as a visual cue (CS). A mouse was placed in a transparent chamber surrounded by screens to present the patterns, with a foot-shock grid floor. We split the chamber into “shock zone” and “safe zone,” notified by distinct visual patterns on the screen. If a mouse failed to leave the shock zone for the period of visual pattern presentation (7s), the animal received a US (0.2mA, 2s). During the task, we recorded dopamine (DA) signals of the tail of striatum (TS) and ventral striatum (VS), with dopamine sensor DA3m using fiber photometry. Dopaminergic responses in the TS clearly reflected the meaning of each pattern. The level of response was stronger when the mouse confronted the “shock delivery” than the “safety” pattern. Furthermore, TS activity remained elevated throughout the CS period in trials where the mouse failed to avoid the shock. Then we added a new “both shock” condition associated with a unique pattern to observe the response to this situation. The activity at the onset of CS was highest at the “both shock” condition but overtaken by that of trials where avoidance failed before the US onset. We demonstrated that mice were able to learn and distinguish the patterns before the CS was terminated. A clear distinction between harmful and safe environments was reflected in the DA activity in TS. The separation between levels of activity became clearer as learning progressed, and the response at the wall-crossing event showed a unique pattern indicating that the mice perceived the outcome of the crossing action.

Keywords: active avoidance, fear memory, augmented reality

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Disrupted Amygdala–Frontal Connectivity Drives Psychopathy-like Antisocial Behavior Following Early Life Social Stress

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Callous and unemotional traits, a hallmark of psychopathy, often accompany persistent antisocial behavior. However, whether such behavior is driven by specific neural circuits modulated by impaired empathy remains unclear. Here, we developed a novel behavioral paradigm in which an observer mouse delivers pain to a conspecific via nosepoking, reflecting psychopathy-like behavior. Using a chronic social stress model, we found that nearly half of the stressed mice engaged in this antisocial behavior. This phenotype was linked to disrupted connectivity between the infralimbic medial prefrontal cortex (IL-mPFC) and the basolateral amygdala (BLA). Circuit-specific optogenetic manipulations were sufficient to either induce or rescue this behavior, suggesting a causal role of BLA-IL circuitry. These findings provide a new framework for modeling empathy-related antisocial traits in rodents and may inform potential circuit-level targets for therapeutic intervention.

Keywords: Psychopathy-like behavior, Stress model, Amygdala–Frontal Connectivity

Acknowledgements:

Genetic and environmental architecture of functional brain networks underlying fluid and crystallized intelligence

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Understanding the neurobiological mechanisms of intelligence is a central goal in cognitive neuroscience. This study used twin-based structural equation modeling with resting-state fMRI to dissociate genetic and environmental contributions to fluid and crystallized intelligence. Multivariate ACE modeling showed that genetic correlations (r_A) were strongly linked to fluid intelligence, identifying 61 associated functional connections (FCs), whereas only two FCs were related to crystallized intelligence. In contrast, environmental correlations (r_E) revealed far more associated FCs, with 2,192 for fluid intelligence and 190 for crystallized intelligence, indicating a prominent role of environmental influences, particularly for crystallized intelligence. Genetically associated FCs were exclusive to fluid intelligence and primarily involved cortex–cortex connections, especially within and between the Default Mode and Control networks. One FC between the right cerebellum and left inferior parietal lobule showed shared genetic association with both intelligence domains. Environmentally associated FCs were widely distributed across networks, with greater involvement of Visual and Somatomotor systems and extensive cortex–subcortex connections for crystallized intelligence.

Functionally, FCs linked to fluid intelligence showed positive associations with performance, whereas crystallized intelligence showed negative correlations, particularly across cortex–subcortex links. Higher global efficiency was positively associated with crystallized intelligence, indicating distinct genetic and environmental brain network architectures.

Keywords: Intelligence, functional connectivity, heritability

Acknowledgements:

Enhancement of hippocampal memory through behaviorally coordinated theta stimulation in the medial septum in rats

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The hippocampus plays a crucial role in spatial navigation by forming internal representations of space through the activity of place cells. Hippocampal theta oscillations (4–12 Hz) are known to support spatial coding of the place cells by temporally organizing neuronal activity during memory encoding. Previous studies have shown that enhancing theta oscillations through medial septum (MS) stimulation can rescue spatial memory deficits in disease models. However, results in healthy animals remain inconsistent, possibly due to non-specific stimulation protocols that fail to align with critical information-processing phases. In this study, we hypothesize that theta oscillations may play a crucial role during the encoding phase, when task-relevant spatial information must be integrated to form new representations. To investigate this, we examined the effects of encoding-phase-specific MS theta stimulation on spatial task performance. Rats performed a goal-directed spatial task on a VR-based linear track. The animals had to stop in a specific reward zone for a certain duration (500 ms) to receive a water reward. The rats relied on allocentric visual cues to locate their current position and estimate the reward zone. During the task, we delivered theta stimulations (7.7Hz; 5 trains of 200Hz, 10 pulses per train) to the MS at reward zone entry, which we consider a critical encoding phase, to enhance hippocampal theta and test its effect on learning. We discovered that theta stimulation facilitated the learning of the reward zone location, reducing days-to-criterion from 7 days to 2.5 days in one rat. To investigate the neuronal mechanisms underlying this learning enhancement, we are developing a setup that allows simultaneous MS stimulation and single-neuron recordings in the hippocampus. This study aims to validate a causal role of hippocampal theta in spatial learning, and further investigate the possibility of learning enhancement by targeting the hippocampal system.

Keywords: Hippocampus, Theta oscillation, Brain stimulation

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Semantic event boundaries enhance episodic memory through cortico-hippocampal event segmentation

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Humans can efficiently remember continuous experiences by segmenting them into discrete, meaningful events. Event boundaries are critical points of segmentation that shape the organization of episodic memory. Prior work has largely focused on event boundaries defined by perceptual or contextual shifts, which have been shown to enhance memory within events while disrupting memory across events. Given the importance of semantic knowledge in human cognition, we investigated how semantic boundaries shape the temporal organization of episodic memory and its cortico-hippocampal correlates. 94 participants encoded ten-word sequences in fMRI. In no-boundary trials, all words belonged to a single semantic cluster defined by Word2Vec embeddings, whereas boundary trials contained a semantic cluster shift midway through the sequence. In a subsequent temporal order judgment task, the presence of semantic boundaries improved memory, particularly for across-boundary judgments. A temporal distance effect, reflecting reduced accuracy for temporally proximal items, did not emerge for across-boundary judgments, indicating that semantic boundaries promote hierarchical event organization. At the neural level, semantic boundaries elicited greater activation in cortical regions engaged in event segmentation, including the lingual gyrus and high-level visual areas. Representational similarity dynamics revealed that crossing a semantic boundary induced abrupt shifts in activation patterns within the hippocampus, reflecting a restructuring of memory representations following boundaries. Similar dynamics were observed in the parahippocampal cortex and orbitofrontal cortex, suggesting a coordinated cortico-hippocampal segmentation of semantic boundaries during episode encoding. Our findings demonstrate that contextual shifts in conceptual space enhances and facilitates the chunking of experience through coordinated modulation of cortico-hippocampal event memory representations.

Keywords: Event boundary, Semantic space, fMRI

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Conjunctive representation of value and space in the retrosplenial cortex

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The granular retrosplenial cortex (gRSC), a hippocampus-connected subdivision of the retrosplenial cortex (RSC), is involved in diverse cognitive functions including spatial navigation, episodic memory, and value-based decision making. To probe the underlying neural mechanisms, we examined the effects of inactivation and neural activity in the gRSC of mice performing a dynamic foraging task on a figure-8-shaped maze. Bilateral inactivation of the gRSC using muscimol severely impaired the animals' choice behavior. Analyses using reinforcement learning models suggest that gRSC inactivation increases choice bias while reducing learning rate. In addition, optogenetic inhibition of the gRSC during the reward-to-delay period reduced the probability of rewarded trials and altered choice outcomes, consistent with a role for the gRSC in subsequent choice behavior. Imaging data revealed that gRSC neurons encoded a range of task-relevant signals, including robust representations of decision value before choice and chosen value after choice, alongside spatial information. Further analysis revealed a positive relationship between decision value signals and spatial information in neurons significantly tuned to decision value. Moreover, gRSC pyramidal neurons were modulated by hippocampal axonal inhibition during the task, implying that hippocampal inputs convey task-related information to the gRSC. These findings support the view that the gRSC integrates value and spatial signals under hippocampal influence to guide flexible decision making. Ongoing work aims to elucidate the circuit-level mechanisms underlying this integration.

Keywords: Retrosplenial cortex, Value-based decision making, Hippocampal-cortical interaction

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Layer-dependent activity in macaque primary visual cortex during apparent motion perception

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Visual processing in the primate cortex depends on interactions between feedforward and feedback pathways, yet how these signals jointly shape perception remains unclear. To address this, we exploited apparent motion-induced masking, a phenomenon in which a physical stimulus is perceptually suppressed by an illusory motion path. Apparent motion (AM) arises between two alternating visual stimuli (inducers) and selectively impairs detection of stimuli whose visual features are similar to the interpolation between the inducers. Two rhesus monkeys performed a task reporting the absence or spatial location of a briefly flashed visual target presented within the apparent motion path. Targets were Gabor patches with multiple orientations and contrast levels, while the inducers had a fixed orientation and contrast. Behavioral performance was high overall but declined when target orientations matched those of the inducers, confirming a robust AM-induced masking effect. To examine the neural basis of this perceptual phenomenon, we recorded activity across all cortical layers of primary visual cortex (V1) using multi-contact laminar electrodes, with laminar boundaries estimated by current source density analysis. In the presence of apparent motion, neuronal responses to the target were reduced relative to the control condition, with suppression increasing progressively as target orientation approached that of the inducer. This orientation-dependent suppression persisted beyond target offset and was most pronounced in the supragranular layer. Consistent with the spiking results, local field potential responses, particularly gamma-band power, were also reduced for targets similar in orientation to the inducer. Together, these findings show that apparent motion produces a sustained, orientation-dependent reduction in sensory responses, most prominently in the supragranular layers, and consistently expressed across both spiking activity and LFP signals.

Keywords: V1, Macaque, Apparent motion

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Anatomical Dissociation within the mPFC for Self- and Other-Focused Social Comparison

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This study examined the neural mechanisms underlying social comparison by integrating goal orientation (self- vs. other-referential) and a regulatory focus (promotion vs. prevention) framework. Forty participants performed a cued response task across five conditions: be faster than self/other (promotion) or not slower than self/other (prevention), along with a control condition with no-feedback.

While behavioral hit-rates showed no significant main or interaction effects, univariate fMRI revealed a robust self-other dissociation within the medial prefrontal cortex (mPFC). The rostral mPFC (rmPFC) was uniquely activated for the Self-Promotion vs. Self-Prevention contrast, whereas the dorsal mPFC (dmPFC) was uniquely recruited for the Other-Promotion vs. Other-Prevention contrast.

Multivariate pattern analysis (IS-RSA) further elucidated how individual behavioral profiles modulate these neural patterns. K-means clustering of hit-rates identified two distinct phenotypes: self-focused and other-focused individuals. By correlating behavioral RDMs with neural RDMs for “Hit” conditions and applying the cluster-strength method, we identified cluster-specific parcels. Specifically, for both Self-Hit and Other-Hit contrasts, self-focused participants showed unique neural representations in vmPFC and pgACC/rmPFC, whereas other-focused participants showed unique representations in the dACC and dmPFC.

These results suggest a clear anatomical and functional dissociation within the mPFC during social comparison. The rmPFC, linked to subjective value and self-relevant integration, specifically encodes self-related outcomes in self-focused individuals. In contrast, the dmPFC, associated with externally oriented evaluative processing, characterizes individuals focused on social-referential outcomes. Collectively, these findings provide evidence for a specialized mPFC architecture that navigates the complex interplay between individual regulatory goals and social comparison.

Keywords: Medial prefrontal cortex(mPFC), Social Comparison, Self-other dissociation

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Attentional Control Induced by Distinct Meditation Practices Differentially Modulates Retrieval-Induced Forgetting

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Retrieval-induced forgetting (RIF) is a phenomenon in which related information about a target memory is lost through repeated selective retrieval. This adaptive process improves focus on goal-relevant data by suppressing competing memories. Inhibition-based theories suggest that RIF occurs during retrieval practice through active suppression of weaker memories. Although attentional control likely influences retrieval competition, few studies have directly tested whether acute selective attention training affects this mechanism. In the present study, we examined how meditation-type-specific attentional control differentially affects RIF. Using a Korean-adapted RIF paradigm, participants encoded category-word pairs and engaged in either focused-attention meditation (FAM) or loving-kindness meditation (LKM), followed by selective retrieval and testing. Results indicated that the reaction time inhibition index was more sensitive to individual factors than the correct rate. Interestingly, higher meditation effects increased the correct rate, whereas personal traits did not significantly account for RIF. Post-hoc analysis showed an increased correct rate but also longer reaction times to self-related information, implying that the retrieval pattern may depend on the relevance of the information. Self-reports revealed that affective changes impacted reaction time. Overall, these findings suggest that selective attention does not universally enhance inhibitory memory processes; rather, the way attention is directed—narrow, target-focused versus broad, non-competitive—dynamically affects retrieval and inhibition. This study emphasizes the importance of considering qualitative aspects of attention in understanding adaptive forgetting.

Keywords: Retrieval-induced forgetting, Selective attention, Meditation

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Visual scene completeness and spatial configuration of landmarks drive CA3 place cell recruitment

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Visual stimuli exert strong control over hippocampal representations, yet the properties triggering initial place cell recruitment remain unclear. In VR, we have shown that CA1 place cells emerge with individual landmarks, whereas CA3 requires a visual ensemble (Shin et al., 2022). Here, we investigate in detail the properties of visual environmental stimuli that underlie the emergence of CA3 spatial selectivity. Single-unit activity was recorded from CA1 and CA3 in Long-Evans rats (n=8) navigating a virtual linear track. In an environment with a black background, discrete landmarks were incrementally increased from 2 to 20. While CA1 place-cell recruitment was robust even with sparse landmarks, CA3 activity remained extremely minimal until an ensemble of 18-20 landmarks triggered an abrupt recruitment of place cells. The activated place cells suddenly acquired spatial selectivity from inactive states, suggesting the presence of an activation threshold for CA3 recruitment. To determine whether CA3 representations reflect landmark density or spatial configuration of landmarks, we swapped landmark positions while preserving density and identity. This manipulation triggered complete remapping in CA3 place cells, confirming the spatial configuration as the primary factor driving CA3 activation. We further tested whether the complexity of background scenes facilitate CA3 recruitment. In a monotonous background featuring distant hills, landmarks triggered CA3 recruitment even in sparse conditions, effectively lowering the activation threshold. A forest-like complex background further maximized recruitment and yielded robust place fields. These findings suggest that CA3 leverages complex visual backgrounds to organize landmarks into a coherent configuration. In sum, unlike CA1, CA3 requires stimulus ensembles and complex background scenes drastically facilitate the spatial activities of place cells.

Keywords: Hippocampus, Place cell, VR environment

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Cortical desensitization underlies empathic fatigue

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While empathy is fundamental for adaptive social behaviors and caregiving, excessive or prolonged exposure to others' suffering can lead to adverse outcomes, including decreased prosocial helping, increased stress, and emotional exhaustion—ultimately resulting in empathic fatigue. This burnout-like state arises from empathic distress, an aversive, self-oriented response that motivates withdrawal from others' suffering. However, the neural mechanisms underlying empathic fatigue remain unknown. Here, we established a repetitive observational fear (OF) paradigm modeling empathic fatigue in mice. Following 7 consecutive days of observing conspecifics receiving footshocks, observer mice exhibited reduced freezing responses, indicating emotional numbing to others' distress, along with a significant decline in prosocial helping, including less comforting (allogrooming) and targeted helping (allolicking) toward distressed partners. Activity-dependent neuronal labeling (TRAP) revealed a significant decrease in Fos-positive neuronal populations within the anterior cingulate cortex (ACC), prelimbic cortex (PL), and basolateral amygdala (BLA) following repetitive observational fear exposure. Importantly, *in vivo* single-cell calcium imaging revealed that ACC neurons exhibited a selective reduction in responsiveness to demonstrators' pain-related cues (squeaks, jumps, and runs) across repeated sessions, suggesting a gradual attenuation of empathic signal encoding over time. Collectively, our findings identify blunted ACC responsiveness as a neural hallmark of empathic fatigue, linking repeated exposure to others' suffering with diminished cortical encoding of distress-related social information. These findings provide a mechanistic framework for understanding how empathic engagement can evolve into emotional exhaustion and caregiving burnout in humans.

Keywords: Empathic fatigue, Anterior cingulate cortex, Observational fear

Acknowledgements:

Delayed source information determines biased retrospective valuation during observational learning

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Social polarization fosters selective exposure and echo chambers, as individuals gravitate toward information that aligns with their preexisting views or ingroup perspectives. Although interventions to reduce biased exposure have been proposed, their effectiveness has been inconsistent. To address this gap, we need to understand how group membership shapes observational learning. Prior work shows that individuals trust ingroup sources more than outgroup. Yet it is unclear whether individuals simply ignore outgroup information or initially learn from all sources and then retrospectively adjust the value of outgroup-derived information. To examine this question, we recruited 124 participants spanning liberal, moderate, and conservative partisanship into a novel observational learning task. Participants first predicted and observed a demonstrator's choice and outcome without knowing the demonstrator's group identity, which was predetermined as ingroup or outgroup and revealed afterward. This delayed disclosure of group membership allowed us to rule out the possibility of reduced attention to outgroup members and instead test whether individuals re-evaluate previously acquired information based on group identity. Using computational modeling, we found that participants increased the value of an option more when the demonstrator's choice aligned with their initial prediction and was rewarded, and decreased it more when the choice did not align and was not rewarded. These positive and negative valuation biases were further amplified for ingroup and outgroup demonstrators, respectively. These findings reveal a computational mechanism by which group membership shapes learning from others, demonstrating that social identity can both enhance and diminish the integration of observed information. Together, our results highlight the importance of accounting for group biases when designing interventions aimed at promoting balanced information processing in polarized environments.

Keywords: Social polarization, Observational learning, Group bias

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Standardized MNI-Guided TMS Demonstrates Functional Equivalence to Individualized T1-Guided TMS

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Background: T1-guided neuronavigation (T1-protocol) is regarded as the gold standard for transcranial magnetic stimulation (TMS) due to its high spatial precision, but acquiring individual T1-weighted MRI scans creates significant logistical and financial burdens. MNI template-based neuronavigation (MNI-protocol) offers a practical, cost-effective alternative. However, the functional equivalence of the MNI-protocol versus the T1-protocol remains empirically unverified, posing a critical trade-off. **Objective:** We aimed to determine whether the MNI-protocol yields comparable targeting efficacy to the T1-protocol, specifically testing their functional equivalence. **Methods:** We employed a novel tri-level approach. First, in the behavioral level, we utilized the Size Congruity Effect (SCE) measure to compare functional efficacy during a number comparison task. Second, in the anatomical level, we used coil-to-cortex distances to compare spatial precision. Third, in the electromagnetic level, we employed electrical field simulations to compare the induced E-field magnitude and distribution between the protocols. The second and third levels were conducted across three cortical regions (right IPS, left DLPFC, and left M1) to assess regional generalizability. **Results:** The MNI-protocol demonstrated functional equivalence to the T1-protocol, evidenced by the statistically indistinguishable behavioral outcomes. Electromagnetic simulations confirmed this equivalence, yielding nearly identical E-field results across all three regions. While the T1-protocol yielded significantly shorter coil-to-cortex distances, this anatomical advantage did not translate into functional differences. **Conclusion:** Both protocols produced comparable behavioral and electromagnetic results. These findings support the MNI-protocol as a valid, practical alternative when individualized MRI is unavailable, enabling scalable and cost-effective neuronavigated TMS across diverse research and clinical settings.

Keywords: TMS, Target localization, Neuronavigation

Acknowledgements:

When to give up: Characterizing disengagement from reward-seeking and punishment-avoidance by a continuous adaptive psychometric approach

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When agents repeatedly encounter inescapable outcomes, adaptive disengagement from ongoing actions naturally emerges. While this behavior has been extensively studied in the context of punishment avoidance, how such behavioral and motivational shifts unfold when both reward-seeking and punishment-avoidance are involved remains poorly understood. In this study, we propose a computational approach to characterize adaptive giving-up decisions during hunting and fleeing in a naturalistic 3D Minecraft environment, capturing disengagement from both reward pursuit and threat avoidance. To induce transitions between engaged and disengaged states, we systematically manipulated participants' histories of success and failure. Using a psychometric curve-fitting method, we accurately predicted each participant's win rate as a function of task difficulty. Because conventional psychometric methods based solely on binary outcomes are not well-suited for our chase-and-flee task, which incorporates time-to-success as a critical behavioral variable, we leveraged continuous-valued data within a Bayesian framework. In this task, difficulty was dynamically adjusted online, enabling rapid and flexible estimation of individual-specific parameters. The resulting parameter estimates correspond to a hazard function commonly employed in reinforcement learning models to quantify perceived environmental risks over time. By explicitly expressing task difficulty as a function of time, our method jointly informs both success probability and trial duration. Together, this task and modeling approach provide a principled means to quantify adaptive disengagement under varying reward and punishment contingencies in naturalistic decision-making contexts, thereby laying the groundwork for future studies that link behavioral disengagement to underlying neural dynamics and computational principles in complex, real-world-like environments.

Keywords: motivation, curve fitting, reinforcement learning

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Learning-induced Representational Reorganization in the Prefrontal Cortex Predicts Long-Term Associative Memory

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Long-term memory persistence has been attributed to consolidation processes that rely on interactions between the hippocampus and the prefrontal cortex. In our previous work, we showed that the reorganization of hippocampal memory representations during learning is closely linked to later retention. However, whether representational changes during learning in the prefrontal cortex contribute to the stabilization of memories over time remains an open question. To address this question, we used functional magnetic resonance imaging (fMRI) to examine how neural representations changed throughout learning and whether these changes predicted memory performance after a one-day delay and a four-week delay. We observed that neural patterns for associated stimulus pairs became progressively more similar in several prefrontal areas, including the ventromedial prefrontal cortex (vmPFC), lateral orbitofrontal cortex (lOFC), dorsolateral prefrontal cortex (dlPFC), anterior superior frontal gyrus (aSFG), and inferior frontal gyrus (IFG), but only for pairs that were remembered after four weeks. This pattern assimilation did not appear for pairs that were forgotten, nor for pairs that were remembered after one day but forgotten by the four-week delay. In addition, classification analyses showed that representational changes in the hippocampus specifically predicted memories preserved after one day, whereas changes in the lOFC predicted which associations would persist over four weeks. Overall, these results support the idea that prefrontal reorganization during learning predicts the formation of long-lasting associative memories that remain beyond four weeks.

Keywords: fMRI, Associative memory, Lateral orbitofrontal cortex

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Contextual Information Improves Accuracy in Time Estimation

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The ability to estimate time is important in many aspects of daily life. Previous studies have shown that context can bias how long events feel, but it is less clear how explicit knowledge about context changes time estimation. In this study, we examined whether people can use contextual information to improve their performance in a time estimation task. Participants were assigned to one of two groups. The informed group received explicit contextual information about the distribution of durations, whereas the uninformed group did not receive any such information. On each trial, participants observed a stimulus of variable duration and then reported their perceived duration on a continuous scale. We modeled errors in duration estimates using a Bayesian generalized linear model to assess the effect of contextual information on temporal bias. Preliminary results suggest that the informed group showed smaller biases and more accurate duration estimates than the uninformed group across a range of intervals. To further understand how contextual information is used during the task, we applied an iterative Bayesian model that updates an internal estimate of duration on a trial-by-trial basis. In this model, participants' estimates are assumed to arise from combining a context-dependent prior (reflecting the expected duration based on the context) with noisy sensory evidence from each trial. By fitting this model, we estimated how strongly each group weighted contextual information relative to the current sensory input. These analyses indicate that the informed group assigned a higher and more stable weight to context-consistent expectations, leading to duration estimates that were systematically closer to the actual durations. Overall, these findings provide preliminary evidence that making contextual information explicit can improve the accuracy of time estimation and help reduce temporal bias.

Keywords: Time estimation, Contextual information, Iterative Bayesian model

Acknowledgements:

Disruption of Social Experience During Early Adolescence Induces Long-term Behavioral and Neural Alterations

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Social experience during adolescence is crucial for shaping adult cognitive abilities, such as social recognition and behavioral flexibility. Despite evident neuronal and behavioral consequences in adulthood by chronic social stress reported, the developmental progression of neuronal alterations induced by adolescent social interactions remains poorly understood. Moreover, mouse social behaviors have been simply quantified using exploration time, which do not capture the dynamic structure of social interactions. To address these limitations, we investigated adolescent social behavior across multiple social contexts and ages using a machine-learning based pose-estimation approach. We categorized social behaviors into two classes: attentive behaviors reflecting awareness of a conspecific, and proactive behaviors reflecting active and direct social engagement. We first observed a sharp decrease in herding behavior at five weeks of age in home-cage settings. During the sociability task, socially isolated (SI) mice exhibited a significant increase in both attentive and proactive behaviors during early adolescence (4 weeks old). We next assessed whether tactile sensory compensation could attenuate SI-induced behavioral alterations. Although modest improvements were observed during early adolescence, these effects were diminished by adulthood. Finally, we performed chronic oxytocin infusion into the medial prefrontal cortex (mPFC) in SI mice to test whether oxytocin signaling could rescue isolation-induced social behavior alterations. We found that oxytocin infusion into the mPFC during early adolescence selectively normalized heightened proactive behaviors in SI mice. In contrast, oxytocin infusion during late adolescence (5 weeks old) did not produce a significant rescue effect. Together, these findings highlight the dynamic changes in adolescent social behaviors and their potential role in shaping long-term social outcomes.

Keywords: Adolescence, Social Behavior, Medial Prefrontal Cortex

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Precision Neuromodulation in Alcohol Use Disorder: Insights from Drinking Motives and Computational markers

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Alcohol craving is a critical predictor of relapse. While non-invasive neuromodulation, such as transcranial photobiomodulation (tPBM) and transcutaneous auricular vagus nerve stimulation (taVNS), has emerged as a promising approach for craving reduction, individual treatment responses vary substantially. Since addiction involves decision-making impairments driven by distinct motivational states—often conceptualized as shifting from reward-seeking (binge/intoxication) to relief-seeking (withdrawal/negative affect)—identifying behavioral markers is essential for precision medicine. A total of 56 participants were randomized into three intervention groups: tPBM-only, taVNS-only, or combined tPBM+taVNS. Assessments were conducted pre- and post-treatment. Behavioral tasks, including the Delay Discounting Task (DDT), Choice under Risk and Ambiguity task (CRA), and Loss Aversion task (LA), were administered to quantify computational decision-making parameters. Additionally, the Penn Alcohol Craving Scale (PACS) and Alcohol Use Disorders Identification Test (AUDIT) were used to assess craving and alcohol-related problem severity, alongside the Drinking Motives Questionnaire (DMQ) to characterize underlying motivational phenotypes. tPBM-based interventions (tPBM-only and tPBM+taVNS) significantly reduced craving from pre- to post-treatment. To define clinically meaningful individual-level change, responders were defined using an RCI-based threshold (PACS decrease >3). Elastic Net analysis identified three computational parameters (α , β , k) and specific drinking motives (Coping, Conformity, Social) as key predictors of treatment response. These results suggest that the efficacy of non-invasive neuromodulation is closely linked to specific computational markers and motivational profiles.

Keywords: Decision-making, Drinking Motivation, Neuromodulation

Acknowledgements:

Corticotropin-Releasing Hormone Neurons in the Central Amygdala Regulate Defensive Strategy Selection and Fear Learning

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Adaptive survival requires animals to select appropriate defensive strategies in response to threat and the central nucleus of the amygdala (CeA) is essential for orchestrating these actions. The CeA is composed of heterogeneous neuronal populations including SST, CRH, PKC δ , Htr2a, and Tac2 cells. Within this neuronal portfolio, the precise function of its corticotropin-releasing hormone (CRH) neurons remains unclear, with previous studies reporting conflicting roles in passive and active defensive responses. This study investigated the role of CRHCeA neurons in both operant and Pavlovian contexts using a platform-based avoidance paradigm and a chasing robot fear conditioning task. Inactivating CRHCeA neurons impaired acquisition and retention of active avoidance without affecting locomotion, shock sensitivity, or anxiety-related behaviour, indicating a specific deficit in learning or executing avoidance. In contrast to expectations of compensatory freezing, CRHCeA silenced mice displayed reduced conditioned freezing in the Pavlovian paradigm, suggesting a broader impairment in fear expression. Together, these results indicate that CRHCeA neurons coordinate context-dependent defensive behaviours, promoting active responses when control over threat is possible and modulating passive freezing under predictive threat cues. My findings highlight CRHCeA neurons as key integrators of threat information, enabling flexible selection of defensive strategies.

Keywords: Central amygdala, Corticotropin-releasing hormone neurons, Active Avoidance

Acknowledgements:

Distinct computational markers of reward learning in anxiety and depression

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Anxiety and depression are highly prevalent mental health disorders that frequently co-occur and share substantial symptom overlap, particularly in altered reward learning. Both conditions are generally associated with deficits in reward processing, marked by reduced responsiveness to gains and increased sensitivity to losses. Discovering computational markers based on reinforcement learning that disentangle anxiety and depression is critical for precise characterization of both their unique and shared pathophysiology. To address this gap, we analyzed clinical survey and behavioral task data from a large-scale sample (N=1,010). Participants completed questionnaires measuring state and trait anxiety (STAI-S, STAI-T), depression (BDI), and performed a modified probabilistic reversal learning task. In the task, two stimuli were presented each trial, drawn from a set of three stimuli. Reward contingencies for each stimulus were reversed three times throughout the session. The task included separate gain and loss conditions that were presented in alternating blocks. To first demonstrate that behavior was driven by outcomes, we performed logistic regression to predict current choices from outcomes of the previous five trials, which revealed significant effects of past rewards, indicating reward-based learning. A multi-level autoregressive model revealed that higher level of state anxiety was predictive of shorter history of outcomes taken into account during learning. This effect was interactive with depression level, reflecting on a complex interaction between depression and anxiety. This result aligns with the anhedonic symptoms of depression, which are characterized by blunted reward sensitivity, and suggests that this blunting may extend to learning from outcomes such as avoiding losses in the loss context. These findings suggest distinct computational profiles for anxiety and depression, influencing reinforcement learning through different mechanisms.

Keywords: reinforcement learning, anxiety, depression

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Dopaminergic Activation Patterns in the Central Amygdala during Active Avoidance

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Active avoidance is an adaptive behavior in which an organism performs a specific action to evade aversive stimuli, a process requiring the integration of both Pavlovian and instrumental learning. While it is known that dopamine (DA) activity modulates active avoidance performance, the function of DA within the central amygdala (CeA) during avoidance behavior remains unclear. This study aimed to investigate the role of DA signaling in the CeA during active avoidance. First, we optogenetically activated dopaminergic neurons in the ventral tegmental area of DAT-Cre mice to confirm DA responses in the CeA. Subsequently, we monitored real-time DA release patterns in the CeA using fiber photometry during a platform-based active avoidance task. These results revealed that DA in the CeA exhibited distinct patterns depending on the behavioral outcomes. DA activity in the CeA increased during the CS in successful avoidance trials, whereas reduced DA release was observed in failed avoidance trials. These signal dynamics suggest that DA release in the CeA is associated with behavior following the presentation of a conditioned stimulus during active avoidance. Optogenetic inhibition or excitation is planned to determine their causal role in successful avoidance behavior.

Keywords: CeA, Dopamine, Active avoidance, Fiber photometry

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Age-dependent upregulation of SELENBP1 in the 5xFAD mice

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Selenium-binding protein 1 (SELENBP1), previously implicated in several neurological and psychiatric disorders, has recently been reported to be altered in the brains of individuals with Alzheimer's disease (AD). However, its cellular specificity and development to AD-related pathology remain unclear. Given the prominent role of microglia in amyloid-driven neuroinflammation and the hippocampal regional vulnerability in early AD, clarifying how SELENBP1 is regulated at both regional and cell-type-specific levels is essential. Here, we examined age- and genotype-dependent changes in Selenbp1 expression in the hippocampus and prefrontal cortex of non-transgenic (non-Tg) and 5xFAD mice at 1.5, 3, and 6 months of age. Western blot analyses revealed robust age-dependent increases in Selenbp1 expression in both regions, with no statistically significant genotype-dependent differences. In contrast, immunofluorescent analyses revealed that Selenbp1 levels were selectively increased in amyloid-vulnerable hippocampal subregions, including the dentate gyrus and dorsal subiculum, where it was highly enriched in microglia and largely absent from neurons or astrocytes. Together, these findings indicate that Selenbp1 elevation under AD-like conditions is region- and cell type-specific, reflecting microglial responses that become apparent only with spatially resolved analysis. SELENBP1 may therefore represent a microglia-enriched molecular signature associated with early amyloid pathology.

Keywords: Alzheimer's disease, Selenbp1, microglia, 5XFAD, amyloid

Acknowledgements:

Neural Encoding of Emotional Valuation in the Medial Prefrontal Cortex

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Emotional evaluation, the assessment of the affective value of environments or stimuli, varies substantially across individuals, yet its neural basis remains poorly understood. We developed a mouse model to investigate how the medial prefrontal cortex (mPFC) contributes to individual differences in emotional valuation under conditions of affective certainty and ambivalence. Mice were placed in a two-chamber apparatus in which one chamber was paired with either a positive stimulus, a negative stimulus, or a combination of both, while the other chamber remained neutral. Preference testing revealed significant inter-individual variability across all conditions. Fine-grained behavioral analyses using pose estimation with DeepLabCut uncovered distinct behavioral strategies associated with each affective context. Fiber photometry recordings of mPFC activity using GCaMP8f showed dynamic modulation at the initiation of transitions toward a chamber depending on the affective value of the target context. Strikingly, these signals were strongest during approaches toward the ambivalent context. Consistent with these population-level signals, miniscope calcium imaging further revealed heterogeneous recruitment of individual mPFC neurons at the movement onset. Together, these findings indicate the mPFC as a key neural substrate for action selection under affective ambiguity and suggest a mechanistic basis for individual differences in emotional evaluation.

Keywords: mPFC, emotion, valuation

Acknowledgements:

Parallel posterior parietal circuits balance flexibility and stability during rapid reversal learning

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To adapt rapidly to changing environments, the brain must resolve a trade-off between flexibly updating stimulus-outcome associations and maintaining stable goal-directed motor execution. Although this flexibility-stability balance is fundamental to cognitive control and is disrupted in multiple neuropsychiatric conditions, its underlying circuit mechanisms remain unclear. Here, we demonstrate that the posterior parietal cortex (PPC) reconciles this trade-off via two parallel projection pathways with dissociable computational roles. Using a within-session reversal learning paradigm combined with projection-specific calcium imaging and causal manipulations, we found that PPC-to-AC neurons flexibly reconstructed stimulus-outcome contingencies and mediated rapid behavioral updating after contingency reversal, whereas PPC-to-IC neurons stably represented goal-directed actions, thereby supporting reliable execution of Go responses. Network modeling further demonstrated that parallel top-down architectures enable more efficient and rapid updating of stimulus-outcome associations than integrated architectures. Together, these findings reveal a double dissociation in PPC output pathways that provides a circuit-level solution for coordinating flexible rule updating with stable motor execution. This work offers a mechanistic and computational framework for understanding how the brain balances flexibility and stability during adaptive behavior.

Keywords: Posterior parietal cortex, Flexibility, reversal learning task

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Investigating real-time decision making under stochastic environment in a naturalistic driving task using inverse reinforcement learning

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Naturalistic tasks offer environments where participants can freely interact with large state and action spaces, but analyzing complex data from naturalistic tasks could be challenging. In our previous study (Lee et al., 2024), we developed a naturalistic real-time driving task - the (static) highway task - and applied an inverse reinforcement learning (IRL) algorithm to infer reward functions from the data. With the reward functions inferred by IRL, we found that IRL rewards reflect the internal reward of individuals and they could serve as indicators of impulsivity. In a follow-up fMRI study, IRL reward trajectories and neural activity showed strong correlations in the reward circuit, which suggests that they may reflect internal value representations in the brain. In the present study, we further extended the task by adding stochastic lane changes of the other non-player vehicles (dynamic highway task). Participants should decide their actions considering increased risk of crash and the reward from speed increment. Participants performed both static and dynamic highway tasks in a counterbalanced order. Behavioral results showed significant reductions in both mean speed and mean reward in dynamic conditions compared to static settings. We further analyzed IRL reward trajectories around two key events: cut-ins and overtaking. Preliminary findings suggest that the IRL reward trajectories may differ across impulsivity levels. High-impulsivity individuals exhibited steeper drops in IRL reward during cut-ins, whereas group differences diminished during overtaking in dynamic conditions. We aim to investigate how individual traits such as impulsivity and anxiety are reflected on decision making under uncertainty and IRL reward trajectories.

Keywords: naturalistic task, Inverse reinforcement learning, impulsivity

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Brain Foundation Model with Multi-Task Masking: Region, Random, and Future Prediction

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To robustly handle missing or corrupted fMRI signals, we propose a brain foundation model capable of reconstructing and forecasting whole-brain dynamics from partial observations. Utilizing spectrogram embeddings and RoPE encoding within an encoder-only transformer architecture, the model employs a multi-masking training strategy involving Random, Region, and Future masking. We utilized the HCP 3T fMRI dataset (REST and TASK) from 2,445 subjects across three age groups (Young Adult, Adult, and Development) to ensure broad lifespan coverage. Our results indicate that spatial reconstruction quality remains robust regardless of the pretraining domain, capturing large-scale spatial topology effectively. In contrast, future forecasting is highly domain-sensitive, benefiting significantly from domain-matched pretraining strategies. Notably, reconstruction performance demonstrated an inverse correlation with age (Development > Young Adult > Adult). This pattern suggests that as individuals accumulate diverse experiences, their brain networks become increasingly differentiated and personalized, thereby reducing the efficacy of population-trained models. Furthermore, attention weight analysis revealed a progressive alignment with effective connectivity (EC) and functional connectivity (FC), suggesting the model learns to represent directional information flow patterns. These findings demonstrate that a single foundation model can jointly perform spatial reconstruction and probabilistic forecasting, offering a potential framework to explain inter-individual differences in cognition and behavior.

Keywords: Brain Foundation Model, fMRI, Multi-Task Masking

Acknowledgements:

Real-Time Individual Sleep Prediction Enabled by Probabilistic Modeling of Wearable Data and Functional Brain Activity Analysis

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Insomnia is a major health concern in modern society. However, sleep drive, which plays a significant role in sleep regulation, is not directly observable, making it hard to intervene effectively. We propose a probabilistic model that estimates individual sleep drive and its uncertainty in real time. The model infers sleep drive using only heart rate, sleep onset time, and wake time, and demonstrates a significant improvement over an existing model that does not consider uncertainty. Using wearable data from both the healthy group and the insomnia group ($N = 32$), cross-validation showed that the model significantly predicted wake time, total sleep time, and wake duration on days not used for training. In addition, we examined the relationship between the model's key parameters and prefrontal brain activity measured during Stroop and N-back tasks using functional near-infrared spectroscopy (fNIRS). The sleep model parameters were significantly correlated with task difficulty-related changes in brain activation, particularly in the orbitofrontal cortex (OFC) and dorsolateral prefrontal cortex (DLPFC). These findings indicate that our model can reveal individual differences in sleep regulation and their neurobiological mechanisms using digital phenotype data collected from wearable devices. Our model may be useful for clinical subtyping of insomnia based on digital phenotypes and personalized sleep interventions.

Keywords: Sleep, Insomnia, Functional imaging, Drift-diffusion model, Digital Phenotype

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Neural Mechanisms Linking Interoception and Subjective Preference: Evidence from EEG and fMRI

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Preference is known to reflect internal states and fluctuate with physiological conditions. This study investigated how interoception relates to both task-relevant and task-irrelevant preference processing, using task-based EEG and fMRI. 39 Participants completed a Movie Rating Task during EEG recording, while 42 performed a Preference-Luminance Decision Task during fMRI scanning. Interoceptive sensitivity was measured via the Heartbeat Counting Task. Hierarchical GLM analysis on EEG data revealed that heartbeat-evoked potentials (HEP) during movie recall were heightened as subjective preference levels increased. This modulatory effect significantly correlated with individual metacognitive interoceptive awareness. Next, hierarchical GLM analysis on fMRI data revealed that the anterior insula (AI) showed greater activation during preference than luminance decision trials and encoded the task-irrelevant preference information during luminance trials. Inter-subject representational similarity analysis (IS-RSA) demonstrated that individuals experiencing a larger interrupting effect from task-irrelevant preference information during luminance trials showed more similar AI activation patterns between preference and luminance incongruent trials, suggesting increased preference signal intrusion. Generalized psychophysiological interactions (gPPI) analyses identified a stronger AI-rostral mPFC (rmPFC) connectivity as a key mechanism. Integrating EEG and fMRI evidence, this study provides converging neural evidence that preference decision involves heightened interoceptive processing and that interoceptive metacognition shapes both the processing and regulation of subjective preference information.

Keywords: Insula, Medial Prefrontal Cortex, Heartbeat-evoked Potential (HEP)

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Cortical responses to heartbeats predict alignment with group consensus in moral and affective preference

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Moral judgments and affective preferences are influenced by social norms and group consensus, yet the mechanisms underlying this alignment remain poorly understood. Emerging accounts suggest that interoception may contribute to the internalization of social norms by providing embodied signals that support value-based evaluations. However, the role of cortical responses to cardiac signals in preference alignment with group consensus has not been fully elucidated. The present study investigated how interoceptive accuracy (IAcc) and heartbeat-evoked-potentials (HEP) contribute to individual differences in moral and affective preference alignment. Forty-one healthy individuals underwent EEG recordings during a resting state, a heartbeat counting task, a moral dilemma task, and a movie preference rating task. Preference alignment was quantified as the distance from group-level consensus for moral judgments and movie ratings. HEPs were time-locked to ECG R-peaks, and statistical significance was assessed using cluster-based permutation tests. Behaviorally, higher IAcc was associated with greater alignment with group consensus in moral judgments but conversely linked to more idiosyncratic choices in movie preferences. Neurally, frontocentral HEP amplitudes were enhanced during the heartbeat counting task and positively correlated with IAcc. Trial-wise analyses indicated that the association between HEP amplitude and preference distance varied as a function of IAcc. This study reveals interoceptive processing's dual role: heightened sensitivity promotes moral conformity while fostering affective differentiation. Neurophysiologically, HEPs link cardiac-brain interactions to these domain-specific alignments. These findings provide a biological basis for how internal physiological signals modulate the balance between social integration and individual identity.

Keywords: Heartbeat-evoked potential, Interoception, Preference

Acknowledgements:

Versatile visual stimulation for awake mice in ultra-high field fMRI reveals diverse patterns in visual and prefrontal cortex

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Functional magnetic resonance imaging (fMRI) enables non-invasive investigation of brain wide neural dynamics. Awake mouse fMRI further allows the study of cognitive processes through manipulation techniques such as optogenetics. However, delivering controlled visual stimuli in awake setups is technically challenging, especially in ultra-high field MRI. To address this, we developed a pixel-level visual stimulation system optimized for awake mouse 15.2T fMRI. Because conventional front-mounted displays cannot fit inside the restricted bore of a 15.2T scanner, we placed a beam projector outside the magnetic field. Positioned 2 m behind the coil to avoid magnetic attraction and signal interference, the projector delivered stimuli to a rear screen through a custom non-reflective tube. This configuration enabled presentation of versatile visual stimuli. Using this system, we mapped whole-brain responses to flicker (5 Hz), moving gratings (0.5, 2, 8 Hz), virtual linear-track scenes, and naturalistic cat stimuli. Flicker evoked the strongest responses across visual areas and uniquely drove activation in the anterior cingulate cortex (ACC). In contrast, moving gratings induced minimal ACC activity but showed clear frequency tuning: visual cortices responded maximally at 2 Hz, whereas the superior colliculus responded slightly more to 0.5 Hz. Additionally, 2 Hz grating, virtual tracks, and cat movies drove similar responses in visual cortices, but ACC activation diverged markedly (grating < linear track < cat). Notably, only the linear track additionally recruited the medial septal nucleus. These findings demonstrate that our platform reliably elicits distinct, stimulus-dependent whole-brain activation patterns. We are extending this platform with reward delivery and behavioral monitoring to link neural activity with behavior in complex visual paradigms involving associative learning, spatial navigation, and cognitive planning in awake mouse fMRI.

Keywords: Awake mouse 15.2T fMRI, Visual stimulation system, Rear projection

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PLC γ 1 Regulates the Transition from Social Novelty to Familiarity in Hippocampal CA2 Ensembles

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Hippocampal CA2 is essential for social memory, yet the mechanisms by which it rapidly updates social representations remain elusive, particularly given its resistance to classical synaptic plasticity. Here, we investigated whether PLC γ -1-dependent signaling engages a distinct neurotrophic signaling pathway to drive the rapid evolution of CA2 population codes. Combining longitudinal in vivo calcium imaging with a two-block social familiarization assay, we tracked CA2 ensemble dynamics in CA2-specific *Plcg1* conditional knockout (cKO) mice as they transitioned from social novelty to familiarity. While control mice showed rapid behavioral habituation accompanied by the emergence of stable, stimulus-specific neuronal ensembles, cKO mice displayed impaired habituation and unstable tuning. Demixed principal component analysis (dPCA) further revealed that in controls, population activity trajectories progressively diverged from a novelty-encoding subspace to a distinct familiarity regime. In contrast, cKO ensembles failed to execute this orthogonalization, remaining dynamically trapped in a state indistinguishable from the initial novelty response. Collectively, our results demonstrate that PLC γ -1 signaling orchestrates the dynamic orthogonalization of CA2 representations, serving as a distinct computational switch that stabilizes familiarity-related representations.

Keywords: Hippocampus, Social memory, in vivo Calcium imaging

Acknowledgements:

Verification of the Mirroring Hypothesis: The Effect of Unstructured Conversation with Chatbots on Linguistic and Emotional Assimilation

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This study investigates the impact of perceiving a conversational partner as an AI on linguistic and emotional assimilation during unstructured interactions with Large Language Models (LLMs). By comparing user responses to an identically prompted ChatGPT-5 model labeled as either a 'Human Participant(C)' or an 'AI', we analyzed the effects of perceived identity on communicative behavior of both AI and humans.

Results revealed that the 'AI' label facilitated higher information extraction accuracy, driven by increased user self-disclosure. However, this condition simultaneously reduced natural assimilation while increasing 'Conversational Load'—manifested as higher linguistic complexity and emotional regulation effort. Notably, the AI labeled as 'Human' induced higher levels of assimilation than actual human-to-human interactions. These findings suggest that while explicit AI perception triggers strategic shifts and increased cognitive load in humans, LLMs possess the latent potential to function as equivalent social partners capable of fostering deep assimilation.

Keywords: Human-AI Interaction (HAI), Mirroring Hypothesis, Unstructured Conversation

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Eye-tracking based Bayesian inference for adaptive decision making and planning processes in a dynamic environment

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Adaptive decision-making is a fundamental aspect of higher cognitive function, requiring the continuous updating of goals and plans in response to environmental change. However, the neurocognitive mechanisms underlying planning remain elusive, as previous studies have largely examined planning in static or single-goal settings, overlooking the dynamic processes of plan construction, evaluation, and revision. Here, we introduce a novel arithmetic task designed to probe multi-step planning and flexible goal switching in a dynamic environment. In this task, participants selected one of two target numbers and formulated plans to reach the chosen target as closely as possible using available component numbers and operators. At each step, these components changed stochastically, requiring participant to decide whether to maintain the current goal or switch to the alternative target. Real-time goal switching was inferred from estimated target utilities, leveraging eye-tracking data within a Bayesian framework. Model likelihoods quantified how gazed items contributed to the expected value of each target given the current state, enabling characterization of individual differences in integrating new information during adaptive decision-making. Behavioral results showed flexible goal adaptation during plan execution, including rapid revision of initial plans when they became infeasible and goal switching as relative utilities changed. Model comparisons further showed that higher performance was associated with decision sequences that converged on plans minimizing the distance to the target. Neural activity in reward processing and valuation networks, tracked utility-based goal-switching probabilities, implicating these regions in adaptive planning. Our findings suggest that adaptive goal switching, reflected in eye gaze patterns, is supported by neurocognitive processes that continuously monitor and update the relative utility of competing goals in a dynamic environment.

Keywords: Eye-tracking, Adaptive planning, fMRI

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Brain's functional connectome patterns during naturalistic movie-watching predict real-life political judgments and decisions

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This study investigated neurobehavioral differences between liberals and conservatives in distributive decision-making and neural processing, and tested whether these differences predict real-world political behavior. We collected neuroimaging, behavioral, and survey data from 128 participants with liberal, conservative, and centrist orientations. From April to October 2024, participants performed a resource allocation task and underwent a movie-watching fMRI, followed by a series of online surveys during the political upheaval of South Korea (December 2024–June 2025), in which participants' political judgments and decisions were assessed at three time points (post-martial law, post-impeachment, and post-presidential election). In the resource allocation task, participants chose whether to keep or share resources with ingroup or outgroup members. To quantify allocation preferences with the recipient's social distance and group membership, choices were fitted with a hyperbolic function to estimate social distance discounting rates based on the recipient's group membership. Using connectome-based predictive modeling (CPM), we identified whole-brain connectivity patterns during neutral, social, and political movie watching that reflected allocation preferences and political orientation and tested whether these patterns predicted real-life political judgments and decisions. Liberals showed inclusive allocation preferences with consistent sharing across social distance and group membership, whereas conservatives showed parochial preferences with steep reductions toward distant and outgroup recipients. Brain connectivity patterns reflecting both allocation preferences and political orientation significantly predicted individuals' political responses. This study identified neurobehavioral markers predicting real-world political responses based on fundamental differences in allocation preferences and neural processing between liberals and conservatives.

Keywords: Political orientation, Allocation preferences, CPM

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Neural Representations of Clearly Experienced Subjective Emotion

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Subjective feeling, the consciously accessible aspect of one's internal emotional experience, is a core component of emotion, alongside physiological and behavioral responses. Yet individuals do not always experience these feelings with the same degree of clarity—some emotions are vividly experienced, whereas others are vague or mixed. This study examined whether the brain represents emotional experiences differently as a function of their subjective clarity. We conducted a functional magnetic resonance imaging (fMRI) experiment in which participants viewed images from the International Affective Picture System (IAPS) and rated the intensity of their subjective experiences of happiness, sadness, and fear for each image. Whole-brain univariate analyses revealed that cortical regions—including the ventral frontal cortex, fusiform gyrus, and anterior temporal areas—were robustly engaged during emotional processing, regardless of whether emotions were experienced clearly or vaguely. We then performed multivariate pattern analysis to test whether neural processing differs between clearly and vaguely experienced emotions. Clear emotional experiences were associated with greater pattern similarity in the right posterior superior temporal sulcus (STS) compared with vague emotional experiences. To assess whether the effect of emotional clarity depends on emotion category, we compared clearly experienced trials of happiness, sadness, and fear. No reliable differences in pattern similarity were detected across emotion types, suggesting that the observed effect in the right posterior STS is more likely driven by emotional clarity than by emotion category. Taken together, these results identify neural correlates of emotional clarity and suggest a prominent role for the right posterior STS in representing the subjective clarity of emotional experience.

Keywords: Superior Temporal Sulcus, Emotional clarity, Human fMRI

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A Neurocomputational Basis for Hysteresis in Pain

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Hysteresis refers to a phenomenon wherein a system's output depends on both current input and prior states. While widely observed in natural systems, including pain perception, its neural implementation remains unclear. Here, we investigated this using an fMRI pain experiment and recurrent neural network. Human participants reported perceived intensity and pain/no-pain decisions under dynamic thermal stimuli, both exhibiting significant hysteresis. The models trained to mimic these behaviors revealed that the degree of hysteresis was driven by two structural features: strengthening self-recurrent weights increased hysteresis, whereas strengthening lateral weights decreased it. This demonstrates that the model implements self-persistence and variability, key features of hysteresis, via these structures. These structural weights were stronger in model units important for hysteretic behaviors, driving high temporal autocorrelation and variance, respectively. Importantly, we found corresponding activity patterns in fMRI data, suggesting the model's structural features serve as a plausible basis for perceptual hysteresis in the brain.

Keywords: fMRI, Hysteresis, RNN

Acknowledgements:

The Role of Voluntary Attention in Cognitive-map Building in the Hippocampus

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Active driving facilitates accurate spatial map formation, whereas passive travel often limits it. This discrepancy implies that acquiring a cognitive map requires voluntary attention (VA) toward allocentric cues. We hypothesized that VA is essential for integrating discrete scenes into a coherent spatial construct in the hippocampus. To test this, we recorded spatial firing patterns of single units simultaneously in the dorsal hippocampus (CA1) and posterior parietal cortex (PPC) of body-fixed rats (n=6) performing a spatial memory task in a virtual reality (VR) environment. Rats navigated a linear track by alternating between running on a treadmill and pushing a lever (>15 trials/condition). Crucially, both methods required VA to allocentric cues despite distinct motor demands. Rats successfully navigated and stopped at the reward zone in both conditions (70% of stops occurred near the reward zone). To analyse spatial firing, we excluded spikes occurring during sharp-wave ripple periods or immobility (VR velocity of 0 cm/s). We defined spatially tuned cells based on significant spatial information scores (> 0.1 , $p < 0.01$). In CA1, spatially tuned cells were observed in all conditions [Running (64%), Lever (15%), and Conjoint (21%)], with neurons maintaining place fields across distinct modalities. However, the proportion of spatially tuned cells in the PPC was significantly lower than in CA1. These results suggest that the hippocampal cognitive map is preserved across different navigational methods, provided the animals utilize VA to perform a goal-directed spatial task.

Keywords: Hippocampus, Voluntary attention, Electrophysiology

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Prefrontal contributions to dopaminergic computations of reward proximity during foraging

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Living with multiple potential goals, we take into account our current status and reward statistics when making decisions. We developed a novel task to model this more realistic reward environment using mice. In a square chamber, an LED cue is positioned at the center of each side, with a water port installed beneath each cue. When one of the four visual cues is randomly activated, a drop of water is dispensed from the port below the activated cue.

As mice learned the association between LED cue and reward delivery, they gradually approached reward with shorter latency. We observed that the latency increased with the mouse's distance to the reward and visual angle of the cue (< 120 deg from the midline). DA responses to cues decreased with distance and angle, suggesting that mice internally estimate time-to-reward ('predicted latency') based on spatial parameters and this process is reflected in dopaminergic responses.

Next, we examined how the competitive context influences DA responses by adding another animal. To estimate which animal may win the competition, we computed the predicted latency for each mouse based on both its own geometric parameters and those of the competitor. This predicted latency reflects each animal's estimated time to reward. Interestingly, when the mouse is closer than the competitor, the DA activity reliably decreases as the predicted latency increases. These results suggest that mice flexibly incorporate both their own and others' physical parameters to compute reward proximity, reflected in adaptive dopaminergic activity.

To test whether the mPFC contributes to computing angle and distance, and consequently influences competitive decision-making, we injected hM4Di into the mPFC and chemogenetically inhibited it using Deschloroclozapine(DCZ). Latency increased following DCZ treatment but stayed constant with saline. CueDA was markedly weakened suggesting that mPFC inhibition disrupts the computation of cue-based reward proximity.

Keywords: foraging, competition, reward, dopamine signaling

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Selective Cognitive-Enhancing Effects of Ampakines in Normal Subjects: A Fear Conditioning-Based Analysis

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Ampakines, a class of cognitive enhancers, have been reported to exert beneficial effects in animal models and clinical populations across a range of neuropsychiatric and neurodegenerative disorders. However, it remains unclear whether these effects generalize beyond disease models to normal conditions. Accordingly, this study investigates the modulatory effects of CX516 and CX717, subtypes of ampakines, on the formation and maintenance of fear memory using fear conditioning paradigms in a normal rodent model. In all experiments, drugs were administered systemically 30 minutes before behavioral testing. Systemic administration of CX516 (10 mg/kg) increased freezing levels on the second day of cued fear conditioning, but not on the subsequent memory test day. In addition, no behavioral changes were observed in immediate or delayed contextual fear conditioning, the spatial object recognition test, or the open field test. An additional experiment examined CX717 using the same behavioral paradigms. CX717 similarly enhanced fear memory in cued fear conditioning, but did not improve object–context associative memory. Consequently, ampakines appear to exert restricted modulation on the neural circuit related to sensory-based associative fear memory, rather than broadly enhancing overall learning and memory abilities in healthy species.

Keywords: Ampakine, Cued Fear conditioning, Contextual Fear conditioning

Acknowledgements:

How do the children with autism perceive the external world?: Cortex-wide mapping of semantic representation and its dimensionality

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Perception forms the foundation of our internal world model, and in Autism Spectrum Conditions (ASC), atypical perceptual processing is a pervasive trait that may cascade into higher-order social and cognitive impairments. However, the neurobiological mechanisms linking these early sensory anomalies to complex cognitive outcomes remain largely unclear. To address this gap, we investigated whole-brain semantic representations in a large developmental cohort (n=428) comprising typically developing children and those with ASC or ADHD. Leveraging advanced "semantic mapping" techniques on naturalistic movie-watching fMRI data, combined with frame-by-frame annotations and gaze prediction via DeepMREye, we generated high-precision, individualized maps of semantic subspaces. We examined these neural representations through the framework of two prominent large-scale brain models: atypical cortical hierarchy and compromised triple networks. Our analyses quantified deviations in ASC neural representations compared to normative developmental trajectories, rigorously testing for specificity against an ADHD control group and reproducibility across different stimuli. Integrating these findings, we propose a developmentally plausible brain-behavior model. This framework elucidates how atypical semantic representations—mediated by alterations in large-scale functional networks—bridge the gap between primary perceptual anomalies and the hallmark social-cognitive challenges characterizing autism.

Keywords: Semantic mapping, Autism, Perception

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Differential contributions of prelimbic and infralimbic cortex to hippocampus-dependent visual context working memory

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Although hippocampus(HP)-medial prefrontal cortex(mPFC) interactions are critical for working memory, their precise dynamics remain elusive. Anatomically within the mPFC, the infralimbic cortex (IL) receives denser iHP projections than the prelimbic cortex(PL), suggesting functional differentiation. Here, we assessed the functional differentiation between the IL and PL in relation to the iHP in a visual scene(context)-based working memory(WM) task in VR, using Long-Evans rats(n=5) running on a treadmill. The task consisted of three phases: sample, delay, and test. First, rats were exposed to one of four visual contexts(Forest, City, Indoor, Mushroom) during the sample phase followed by a 3-sec delay in a gray-walled tunnel. After the delay, correct lick responses(left or right) were rewarded with honey water(120 trials/session). To test iHP-mPFC involvement, we inactivated the iHP with muscimol(0.3–0.5 μ L/site) while simultaneously recording single-unit activity in PL and IL. iHP inactivation significantly reduced WM accuracy(86% to 62%, $p < 0.001$), supporting its contribution to context-dependent WM. Preliminary analysis indicated that mPFC neural activity was modulated by VR context and choice, with modulation patterns differing across task phases in PL(n=49) and IL(n=39). During the sample phase, iHP inactivation reduced context-driven activity in the PL neurons($p < 0.01$), while it did the same in the IL for both VR context($p < 0.05$) and choice response($p < 0.01$). Choice-related IL activity was further reduced during the delay($p < 0.001$). At the population level, 9.5% of task-modulated IL neurons represented choice across sample and delay phases, a pattern not observed in the PL. Conversely, 13.3% of PL neurons encoded VR context during the sample phase and choice responses during the delay phase. These findings suggest that the PL may encode context while the IL maintains choice-related information across time

Keywords: medial prefrontal cortex, intermediate hippocampus, working memory

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Investigating Cardiac and Respiratory Interoception in Smokers: A Computational Modeling Approach

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Interoception, the perception of internal bodily states, has gained increasing recognition as a mechanistic factor in the etiology and maintenance of substance use disorders. This study investigated cardiac and respiratory interoception in smokers by employing multiple tasks across both domains while controlling for smoking status. Smokers (n=55) and nonsmokers (n=55) completed three interoceptive tasks under both satiated and 12-hour deprived conditions: a heartbeat tapping task, a heart rate discrimination task, and a resistive load respiratory sensitivity task. Computational modeling using Bayesian inference identified latent interoceptive processes, including interoceptive precision, prior beliefs, perceptual sensitivity, and metacognitive awareness. Acute nicotine deprivation distinctly affected cardiac interoception across multiple processing levels. In the heartbeat tapping task, deprived smokers exhibited hyperprecise prior beliefs during guess trials and failed to flexibly update precision weighting during breath-hold trials, indicating altered top-down expectations alongside impaired dynamic precision modulation. In the heart rate discrimination task, deprived smokers showed a perceptual bias toward perceiving faster heart rates, while satiated smokers demonstrated reduced metacognitive sensitivity. These findings suggest that smoking affects both low-level perceptual processes and higher-order metacognitive monitoring. In contrast, respiratory interoception showed no significant differences between groups or states, demonstrating modality-specificity of nicotine's effects. These findings reveal that interoceptive alterations in substance use are hierarchically organized, modality-specific, and state-dependent, necessitating multi-dimensional assessment approaches grounded in computational theory.

Keywords: Interoception, Smoking, Addiction

Acknowledgements:

Sex-Specific Metabolic Control of Depression via FAAH–Endocannabinoid Signaling

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Depression is more prevalent in females than in males, yet the metabolic mechanisms underlying this sex difference remain poorly understood. Using the sub-chronic variable stress (SCVS) model, which induces stress susceptibility in females, we performed non-targeted metabolomic analyses and found elevated levels of arachidonic acid (AA) in the plasma of stressed females (FS) but reduced levels in stressed males. However, exogenous AA administration failed to alter behavior in females, suggesting that the downstream metabolic pathway, rather than AA itself, drives stress susceptibility. We therefore targeted key enzymes involved in AA production by inhibiting fatty acid amide hydrolase (FAAH) and phospholipase A2 (PLA2) using URB597 and quinacrine, respectively. FAAH inhibition during stress selectively alleviated depressive-like behaviors in females, whereas PLA2 inhibition had no effect. FAAH inhibition also normalized stress-induced hyperactivity in the basolateral amygdala (BLA), driven by an excitation-biased excitation/inhibition (E/I) imbalance. These effects were blocked by a CB1 receptor antagonist AM251, indicating that URB597 acts via CB1 receptor activation, likely through increased anandamide levels. Local infusion of URB597 into the BLA recapitulated the behavioral and circuit-level rescue. Consistent with the effects observed following intraperitoneal administration, intra-BLA Together, these findings suggest that metabolic regulation of endocannabinoid signaling in the BLA as a mechanistic contributor to female-biased depression and suggest FAAH inhibition as a potential female-specific therapeutic strategy.

Keywords: Sex difference of depression, Basolateral amygdala(BLA), Endocannabinoid(eCB)

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Forecasting Alcohol Craving via Smartphone-Based Multimodal Data: A Hierarchical Bayesian Machine Learning Approach

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Background: Alcohol craving is a key factor contributing to persistent drinking and relapse after abstinence or treatment. Accurate prediction of craving is essential for early intervention and mitigating the negative impacts of alcohol use disorders. Recent advances in digital phenotyping via smartphone applications enable real-time, continuous behavioral data collection, offering new avenues for craving prediction. However, the adequacy of app data alone remains unclear. Recent studies have explored integrating multimodal data—including neurophysiological and psychological measures—using machine learning (ML) approaches. Taken together, we aim to predict daily alcohol cravings using multimodal datasets. **Methods:** This study evaluates the predictive performance of hierarchical Bayesian multimodal ML models relative to unimodal models trained solely on digital phenotypes. Sixty-one participants were monitored over a 28-day period via a smartphone application that recorded time-series data on mood, craving, sleep, and app usage. Baseline data included electroencephalography (EEG) connectivity, heart rate variability (HRV), and psychological assessment scores. An early fusion strategy was employed to integrate baseline and time-series features, and daily alcohol craving was predicted using a hierarchical Bayesian multimodal machine learning framework. **Preliminary Results:** The unimodal model achieved 0.4057 accuracy, while the multimodal model achieved 0.5020 accuracy, showing improved predictive performance with additional modalities. **Conclusion:** Multimodal ML approaches that combine smartphone-based behavioral data with physiological and psychological measures improve craving prediction. These models may enable individualized monitoring and timely intervention for those at risk of problematic alcohol use.

Keywords: Alcohol Craving, Multimodal, Bayesian Modeling

Acknowledgements:

Neural representations of dyadic social interactions during naturalistic conversations

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Social interactions are dynamic and embedded within rich contexts. However, previous studies of social cognition have often isolated social cues, overlooking prior interactions in inference during naturalistic social exchanges. This study examined 1) whether the brain represents dyadic interactions within a joint interaction space, rather than through independent processing of each individual's actions, and 2) whether such representations support context-dependent inference of others' affect during naturalistic conversation. Participants viewed movies in the scanner (3T, N = 41) depicting dialogue between former romantic partners. We constructed a 2-D dyadic interaction space model, in which each dimension represents one character's avoid-approach intention toward their partner, based on ratings of conversational scenes by independent participant groups (N = 14, 12). This interaction space captured distinct social dynamics, such as mismatched (e.g., rejection vs. pursuit) and aligned intentions (e.g., mutual interest vs. mutual disinterest), forming the basis for joint social interaction features. Additional groups (N = 15, 16) predicted the target character's upcoming valence and then reported experienced valence respectively before and after each scene. We quantified valence prediction updates as the change between predicted and observed valence following a partner's action. Leveraging the joint interaction features, we fit a banded ridge regression model to predict both behavioral and neural responses during movie viewing. The joint dyadic interaction model better explained both valence prediction updates and neural activity in the superior temporal sulcus (STS), outperforming models based on individual intentions alone. These findings suggest that individuals' relational intentions together provide a representational basis for contextualizing complex social interactions and supporting nuanced affective state inference during naturalistic conversations.

Keywords: Social interaction, Prediction, Naturalistic conversation

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Human intracranial correlates of age-related decline in boundary-dependent hippocampal spatial memory

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Boundaries provide stable cues that help anchor spatial representations in navigation. Past studies have reported that a reduced utilization of environmental boundaries contributes to the age-related decline in spatial memory. Given the well-established role of hippocampal theta oscillations in spatial representations, intracranial EEG (iEEG) recordings may reveal their specific role in translating boundary information into spatial memory and the changes that occur in aging. In this study, we analyzed iEEG recordings from 69 presurgical epilepsy patients (ages 19–61) while they performed a computer-based navigation task. Participants were passively driven to a target location and then asked to freely return to that location after being teleported to a random starting point in a virtual rectangular arena defined by wall boundaries beyond which distal landmarks were visible. Specifically, we examined the association between hippocampal theta power and the influence of boundaries on spatial memory performance. We found that age-related spatial memory decline was particularly severe when targets were located near boundaries. This reduced utilization of spatial boundaries was associated with weakened hippocampal boundary representation in older adults. Specifically, whereas hippocampal theta power in younger adults was robustly modulated by distance to the nearest boundary, this relationship was weaker in older individuals. Moreover, older adults showed a bias toward their previous responses, consistent with proactive interference, which led them to respond farther away from boundaries on average. These findings suggest that age-related deficits in boundary-based navigation arise from diminished engagement of hippocampal spatial representations. The observed behavioral patterns further point to a shift toward more generalized, less precise memory retrieval in aging, consistent with experience-dependent overgeneralization in cognitive aging.

Keywords: Spatial boundary, hippocampal theta, aging

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Naturalistic Decision-Making Reveals Exploit Bias and Cost Insensitivity in Abstinent Smokers

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Traditional laboratory tasks such as the multi-armed bandit (MAB) paradigm have been valuable for studying the explore–exploit trade-off but typically simplify decision contexts and exclude real-world costs such as spatial or temporal effort. Understanding how people navigate such costs is crucial in addiction, where exploitative decision biases are well documented but remain understudied in spatially grounded environments. We developed a 3D reinforcement learning task in a Minecraft-based environment, extending the restless MAB paradigm with spatial costs and real-time (10 Hz) tracking of locomotion and gaze-derived indices (e.g., movement and visual entropy). We compared non-smokers (N=36), abstinent smokers (N=24), and satiated smokers (N=23), controlling for abstinence duration. We fit a hierarchical Bayesian Kalman filter with a softmax rule, estimating inverse temperature (β), decay (λ), and a cost parameter (γ) across linear, hyperbolic, and exponential cost models. We hypothesized stronger group differences in the 3D than 2D task, particularly in abstinent smokers, with decision type interacting with locomotion and gaze patterns. In the 2D task, no group differences were observed in β , whereas satiated smokers showed lower λ than non-smokers. In contrast, in the 3D task abstinent smokers showed higher β , lower λ , and lower γ , indicating stronger exploit bias, reduced value updating, and lower cost sensitivity. Locomotion and gaze indices confirmed that explore–exploit contrasts were largest in abstinent smokers. By combining immersive task design and computational modeling, our naturalistic 3D task captured richer individual differences and improved ecological validity for studying addiction-related decision-making.

Keywords: Addiction, Exploration-Exploitation, Decision-making

Acknowledgements:

Planning for Active Sensing

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Intuitively, real-time observation depends on subsequent actions. For optimal active sensing, one must mentally simulate all possible post-observation situations and choose the observation that maximizes the value of later actions. However, cognitive constraints and individual preferences can lead people to rely on simpler heuristics. Here, we introduce a multi-step maze task that distinguishes a mental-simulation planning strategy from simpler heuristics. We also design a hidden Markov model (HMM) to capture the use of multiple sensing strategies and latent switches between them. The results show that eleven out of fifteen participants used at least two strategies, and thirteen employed the planning strategy. We plan to extend our study to large-scale dynamic tasks that require deeper planning. Furthermore, the model's posterior over planning versus heuristic strategies will be used to study the brain activity differences between complex mental processing and mere reward seeking during active sensing.

Keywords: Sensing, Attention, Decision making

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Representation of Spatial Boundaries within Retrosplenial Memory Engrams

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The retrosplenial cortex (RSC) is crucial for integrating spatial information with contextual memory. While recent studies have identified Egocentric Boundary Cells (EBCs) and Boundary Cells (BCs) in the RSC, their specific contribution to memory engrams and retrieval behavior remains elusive. Here, we investigated the functional identity of RSC engram neurons using a cFos-based activity-dependent tagging system combined with in vivo calcium imaging in C57BL/6J mice. We tagged neurons activated during contextual fear conditioning and tracked their activity during a memory recall session. We found that a substantial proportion of these engrams were spatially tuned, identified as EBCs and BCs. During memory retrieval, these spatial engram cells exhibited selective activity patterns corresponding to distinct behavioral states. These findings suggest that the RSC integrates egocentric spatial information into the memory.

Keywords: RSC, Engram, Egocentric boundary cell

Acknowledgements:

Ketamine treatment rescues memory impairments induced by uncontrollable stress

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Major Depressive Disorder (MDD) is a prevalent and debilitating psychiatric condition characterized by persistent sadness, hopelessness, and loss of interest. While conventional antidepressants such as selective serotonin and norepinephrine reuptake inhibitors (SSRIs/SNRIs) are widely prescribed, they typically require prolonged administration (weeks to months) to achieve therapeutic effects, and a significant proportion of patients fail to respond adequately. Recently, ketamine, an N-methyl-D-aspartate receptor (NMDAR) antagonist, has gained considerable attention for its rapid and sustained antidepressant effects, even after a single dose. Given that stress is a major precipitating factor in depression and is commonly used to induce depressive-like behaviors in animal models, understanding the interaction between ketamine and stress is critical. Research into the mechanisms underlying ketamine's antidepressant effects is extensive. Yet, the association between ketamine and stress remains poorly understood. In this study, we investigated the effects of ketamine on stress-induced cognitive impairment in rats. Based on our previous findings, we employed a 60-minute restraint combined with 60 intermittent tail shocks (termed "60-min stress") to induce cognitive impairment. Cognitive performance was assessed using the novel object recognition (NOR) task. Control rats administered saline (10 mg/kg, i.v.) exhibited significant memory impairment following the stress protocol. In contrast, ketamine-treated rats did not show memory impairments, suggesting a protective effect against stress-induced cognitive dysfunction. These results indicate that intravenous ketamine administration effectively reverses stress-induced memory impairment and may offer therapeutic potential for treating cognitive deficits associated with depression.

Keywords: stress, ketamine, working memory

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The anxiolytic impact of exogenous ketone supplements in healthy rats

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Ketones, as an alternative energy source to glucose for the body and brain, have been extensively studied across various disease models in both humans and rodents. Beyond therapeutic benefits in pathological models, ketones have shown an anxiolytic effect. However, most studies have utilized forced-feeding or chronic administration protocols, which may induce confounding stress-related side effects. To isolate the anxiolytic impact of ketones, this study employed a naturalistic ketogenic diet in a healthy rat model. Sprague-Dawley rats were provided ad libitum access to a specialized chow enriched with ketone salts and MCT oil. Blood β -hydroxybutyrate (BHB) and glucose levels were monitored, and behavioral tests were initiated once subjects achieved ketosis (BHB > 0.5 mmol/L). In the open field test and elevated plus maze, the ketone group spent significantly more time in the center zone and open arms, respectively, compared to the standard chow control, without alteration in locomotor activity. In contrast, no significant differences were observed in contextual fear conditioning, forced swim test, and Y-maze. In conclusion, exogenous ketone supplementation showed an anxiolytic effect while having no significant impact on fear memory, depressive-like behavior, or spatial working memory.

Keywords: ketone, glucose, anxiety

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Stress responsiveness shapes neuronal dynamics in the lateral habenula

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The lateral habenula (LHb) plays a central role in stress-related behavioral regulation, yet how stress exposure alters the functional properties of stress-responsive LHb neurons remains poorly understood. Here, we investigated how stress intensity influences neuronal dynamics and behavioral relevance within stress-responsive LHb populations. Using activity-dependent tagging strategies, we identified LHb neurons activated during stress and characterized their physiological properties. Electrophysiological analyses revealed that neurons recruited by excessive stress exhibited enhanced burst firing, increased intrinsic excitability, and elevated presynaptic release probability compared to non-responsive neurons. Fiber photometry recordings further demonstrated stress-related alterations in LHb population activity dynamics in vivo. To examine functional relevance, we optogenetically activated LHb neurons initially recruited during mild stress. Activation of these neurons did not produce significant behavioral effects under baseline conditions. In contrast, following exposure to an acute learned helplessness paradigm, activation of the same neuronal population induced enhanced aversion in real-time place aversion assays and increased immobility in the tail suspension test. Together, these findings suggest that stress responsiveness is associated with dynamic changes in LHb neuronal properties, and that excessive stress alters the physiological and behavioral output of stress-responsive neurons. These alterations may contribute to maladaptive behavioral responses following excessive stress.

Keywords: Stress, Lateral habenula, Electrophysiology

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Network-wise tACS manipulates behavioral performance during coherent motion perception

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Transcranial alternating current stimulation (tACS) is a promising noninvasive tool for cognitive neuromodulation, but its effects on coherent motion perception remain unclear. Pinpointing the electrophysiological patterns supporting the perception of moving stimulus direction—especially with varying motion coherence—is crucial for understanding human attention mechanisms and coherent motion processing. We investigated the differential neuromodulatory effects of tACS on behavioral responses. Participants performed a random-dot kinematogram (RDK) task under three motion coherences (25%, 50%, and 75% coherence). TACS was targeted to either an anterior network (dACC and right anterior insula) or a posterior network (superior parietal lobule and right visual cortex), and individualized EEG oscillatory activity was analyzed within theta and alpha bands.

Participants in the difficult (25% coherence) condition were divided into two subgroups based on their pre-tACS performance accuracy. Poor-performers showed significantly reduced reaction times for both anterior ($Z = 2.242$, $p < 0.05$, pre-tACS: 1649.27 ms, post-tACS: 1401.47 ms) and posterior tACS ($Z = 2.650$, $p < 0.05$, post-tACS: 1339.26 ms), with pronounced improvements of performance accuracy (anterior: $Z = 2.548$, $p < 0.01$, pre-tACS: 72.38%, post-tACS: 85.47%; posterior: $Z = 2.752$, $p < 0.01$, post-tACS: 86.30%). Good-performers exhibited significantly reduced response times only during anterior tACS ($Z = 2.178$, $p < 0.05$, pre-tACS: 1310.61 ms, post-tACS: 1168.84 ms), with no significant accuracy changes.

Our data demonstrate that network-wise tACS modulates behavioral performance during coherent motion perception, with significant effects at the difficult condition. Our findings also suggest that a pre-tACS performance level is a crucial factor to consider when modulating cognitive function using noninvasive neuromodulation techniques.

Keywords: Noninvasive neuromodulation, tACS, Coherent motion perception

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Sex-Dependent Differences in Medial Prefrontal Cortex Activation During Stress Coping

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Sex differences in depression are well established, yet how individual prefrontal neurons encode stress-related behaviors across sexes remains unclear. We used a sub-chronic variable stress (SCVS) model combined with in vivo calcium imaging to examine single-neuron dynamics in the medial prefrontal cortex (mPFC). Under homecage conditions, behavioral profiles and neuron–behavior coupling were largely similar between males and females. After stress exposure, open field behavior did not differ by sex. In contrast, ensemble activity of mPFC neurons revealed robust sex differences: males exhibited higher-amplitude and more sustained calcium responses during center entry, rearing, sniffing, and freezing. These results indicate sex-dependent recruitment of mPFC neuronal ensembles during stress-related behaviors, despite comparable behavioral expression, highlighting divergent neural encoding strategies at the single-neuron level.

Keywords: mPFC, Stress, sex-difference

Acknowledgements: RS-2022-NR072375

Cross-Cultural Validation of Rapid and Reliable Behavioral Paradigms Using Adaptive Design Optimization

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Addiction is a highly heterogeneous condition, with clinical and neurocognitive profiles varying significantly across individuals. This variability underscores the need for reliable and efficient assessment tools that can capture individual differences. While recent neuroscience-based frameworks propose comprehensive, multidimensional batteries, their lengthy administration and low reliability often limit real-world feasibility and clinical utility. To address these challenges, we utilized Adaptive Design Optimization (ADO)—a model-based Bayesian search algorithm that dynamically selects the most informative task designs for each participant. This method enables reliable parameter estimation with fewer trials compared to traditional methods, supporting the development of rapid yet reliable assessments. In this study, we conducted a preliminary validation of ADO-based versions of two key tasks—Delay Discounting (DD) and Choice under Risk and Ambiguity (CRA)—in a non-clinical sample of 119 undergraduate students. Data were collected across four international sites in Bulgaria, the USA, and South Korea. We evaluated both within-visit and between-visit (one-week interval) test-retest reliability using Intraclass Correlation Coefficients (ICC). The results demonstrated that ADO-based tasks improved efficiency and reliability compared to conventional fixed designs, achieving stable individual-level parameter estimates in fewer trials. This study provides a foundation for applying these scalable, ADO-based assessments to clinical populations, including those with substance use disorders, across diverse global settings.

Keywords:

Acknowledgements:

Neural Inertia: Delayed Hierarchical Event Segmentation in Autism Spectrum Disorder

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The brain segments sensory information into hierarchical units to navigate dynamic environments. In Autism Spectrum Disorder (ASD), Weak Central Coherence (WCC) disrupts global integration. We propose this fragmentation stems from impaired neural event segmentation—a failure to register context shifts termed “Neural Inertia.” We hypothesize that this resistance to state updates manifests as prolonged neural state durations, preventing timely scene construction and driving behavioral “insistence on sameness.” We analyzed movie-fMRI from 153 children (95 ASD, 58 TD) from the Healthy Brain Network. Following hyperalignment and HRF deconvolution, we used Greedy State Boundary Search (GSBS) via searchlight to identify neural states. Group differences in median durations were assessed using non-parametric permutation tests (1,000 iterations) and bootstrap resampling to estimate 95% confidence intervals. ASD participants showed significantly prolonged global state durations ($p < 0.0001$). A “sensory bottleneck” was identified in piriform and peri-entorhinal cortices, where transitions were drastically delayed (ASD: 28s vs. TD: 2s), hindering transient cue encoding. Concurrently, the Default Mode Network exhibited a functional dissociation: the Posterior Cingulate Cortex (PCC) showed fragmented states (ASD: 22s vs. TD: 30s), while the Lateral Temporal Cortex (LTC) exhibited extreme inertia (49s). Neural Inertia provides a temporal perspective on ASD neurobiology beyond static connectivity. This dynamic distortion, where states fail to synchronize with changing environments, manifests as a hierarchical shift. Sensory bottlenecks in the piriform cortex may contribute to downstream dissociation—impairing context synthesis in the PCC while over-stabilizing LTC representations. This mechanistic cascade suggests the autistic brain is functionally rewired to prioritize local details, explaining why individuals with ASD may remain detached from dynamic environmental changes.

Keywords: Event segmentation, Autism Spectrum Disorder

Acknowledgements:

Temporally dynamic information processing in the prefrontal and parietal cortices during goal-dependent memory retrieval

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Retrieving goal-relevant information from memory is crucial for adaptive functioning in everyday life. The prefrontal and parietal cortices are known to support this process by orchestrating and flexibly controlling goal-dependent retrieval. However, it remains unclear what types of information these regions encode during retrieval, and whether such processing is temporally dynamic. In this study, we conducted an event-related functional magnetic resonance imaging (fMRI) experiment to investigate the nature and temporal dynamics of information processing in the prefrontal and parietal cortices during goal-dependent retrieval. Participants performed a selective retrieval task, in which they retrieved a cued object from previously learned real-world scenes containing multiple objects. This task was followed by separate object and scene perception tasks. A direct comparison of neural response patterns during selective retrieval and perception revealed transient representations of non-targeted objects in the dorsolateral prefrontal cortex (dlPFC), and of targeted objects in the angular gyrus (AG) and superior parietal lobule (SPL) during the early retrieval phase. During the later phase, although significant overlap between retrieval and perception patterns was not observed, retrieval cue-specific representations were consistently maintained in the dlPFC, AG and SPL. Furthermore, pattern similarity analysis across adjacent time points during retrieval revealed dynamic changes in neural representations during the early phase, followed by stable and temporally consistent representations in the later phase, indicating distinct neural processes across retrieval stages in these frontoparietal regions. Such temporal dynamics were not observed in sensory and hippocampal regions. Taken together, these findings suggest that goal-dependent memory retrieval engages temporally dynamic information processing in the prefrontal and parietal cortices.

Keywords: Prefrontal cortex, Memory retrieval, Functional magnetic resonance imaging (fMRI)

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A dopamine-mediated formation of memory module in the nucleus accumbens for goal-directed navigation

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Spatial memories guide navigation efficiently toward a desired destination. However, the neuronal and circuit mechanisms that underlie the encoding of goal locations and its translation into goal-directed navigation remain unclear. Here, we show in mice that optogenetically induced phasic dopamine (DA) signals to the nucleus accumbens (NAc) are sufficient to form a memory of the desired goal location that directs spatial navigation under threat. During experiences that occur at the goal location, a shelter, converging glutamatergic and dopaminergic inputs to the NAc mediate the formation of this goal-related memory. This memory is represented with shelter experience within a subpopulation of NAc neurons that emerge with shelter experience. Artificial coactivation of the goal-related NAc ensemble and neurons in the dorsal periaqueductal gray (dPAG) sufficiently led to memory-guided escape behavior. These findings provide causal evidence of cognitive circuit modules linking memory with goal-directed action.

Keywords: the nucleus accumbens, episodic memory, goal-directed navigation

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When Expectations Fail: The Role of Predictive Validity in Occluded Duration Estimation

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Accurate estimation of occluded duration (OD) is critical for anticipating dynamic events, such as predicting when an occluded vehicle will re-emerge at a crosswalk. Yet it remains unclear how violations of kinematic expectations distort OD estimation. Here, we investigated the impact of expectation violations on OD estimation using a novel temporal discrimination task. Participants viewed a moving target that traversed an occlusion zone and judged whether the OD was closer to a short or long reference duration. Although the physical OD was held constant across conditions, target velocity was manipulated (accelerated or decelerated) pre-occlusion, post-reappearance, or during both phases relative to a control condition. These manipulations produced three violation conditions: deviation in expected velocity, expected reappearance time, or both. Given that expectation violations are known to trigger pupil dilation, pupillometry was employed to index physiological responses to these violations. Behavioral results indicated that all violation conditions induced larger biases in OD estimation compared to the control, suggesting a strong reliance on top-down processes modulated by the validity of kinematic expectations. Furthermore, pupil dilation was significantly greater when the expected reappearance time or both velocity and time expectations were violated, indicating a physiological marker of prediction error. Finally, a correlation analysis revealed that participants with higher self-reported self-awareness exhibited larger temporal biases. Collectively, these findings demonstrate that expectation violations distort time estimation, highlighting the interplay between predictive processing, physiological arousal, and individual differences in temporal estimation.

Keywords: Occluded duration estimation, Expectation, Pupil dilation

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Distributed neural dynamics underlying sensory-to-motor transformation during olfactory decision making

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Perceptual decision-making requires the transformation of sensory inputs into motor outputs through intermediate processes such as working memory and action planning. While higher-order areas are often emphasized in these processes, how these functions emerge across the brain during learning remains unclear. To address this, we combined acute and chronic Neuropixels recordings in mice performing an olfactory delayed match-to-sample task. In the acute experiments, we recorded dozens of brain areas spanning the full olfactory decision-making pathway—from early sensory regions to hippocampal, frontal, and motor areas. Early sensory areas such as the anterior olfactory nucleus (AON) showed robust delay-period odor-selective activity and match/non-match category, dissociable from licking direction. In contrast, anterior-lateral motor cortex (ALM) encoded licking direction but not perceptual decision, while orbitofrontal (OFC) and prefrontal (PFC) cortices carried mixed decision and motor representations. To examine how these representations develop with learning, we performed chronic Neuropixels recordings in AON, OFC, and ALM, tracking the same neurons across weeks. Odor-selective delay activity in AON increased with learning, and categorical decision making signals became more prominent and stable. These learning-related changes were also observed in downstream areas, suggesting that task-relevant signals are increasingly transferred and utilized across regions. Together, our findings reveal a learning-dependent reorganization of decision-related neural activity across distributed brain regions, where early sensory areas not only maintain sensory information but also serve as critical hubs for initiating flexible, experience-driven sensorimotor transformations. This challenges classical hierarchical views and highlights the adaptive nature of brain-wide networks in linking sensation to action.

Keywords: Neuropixels, Decision-making, Electrophysiology

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